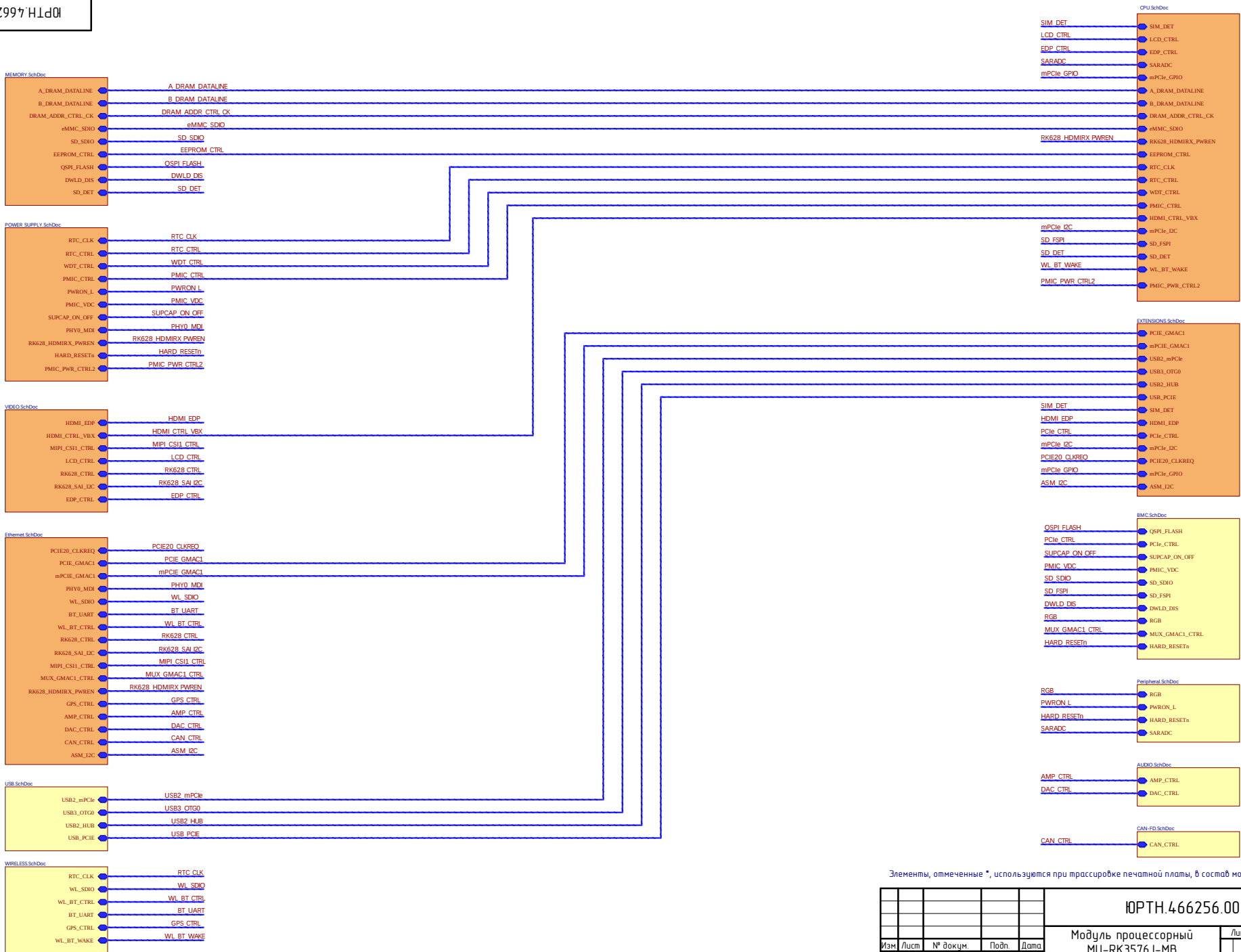
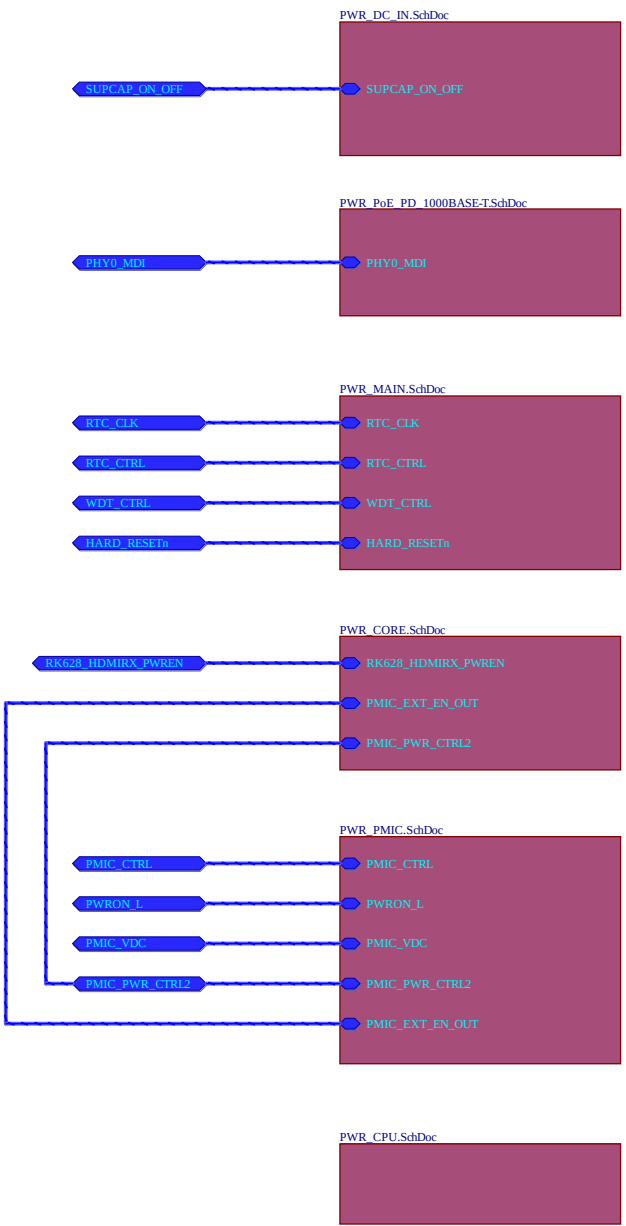




Элементы, отмеченные *, используются при трассировке печатной платы, в состав модуля не включаются.

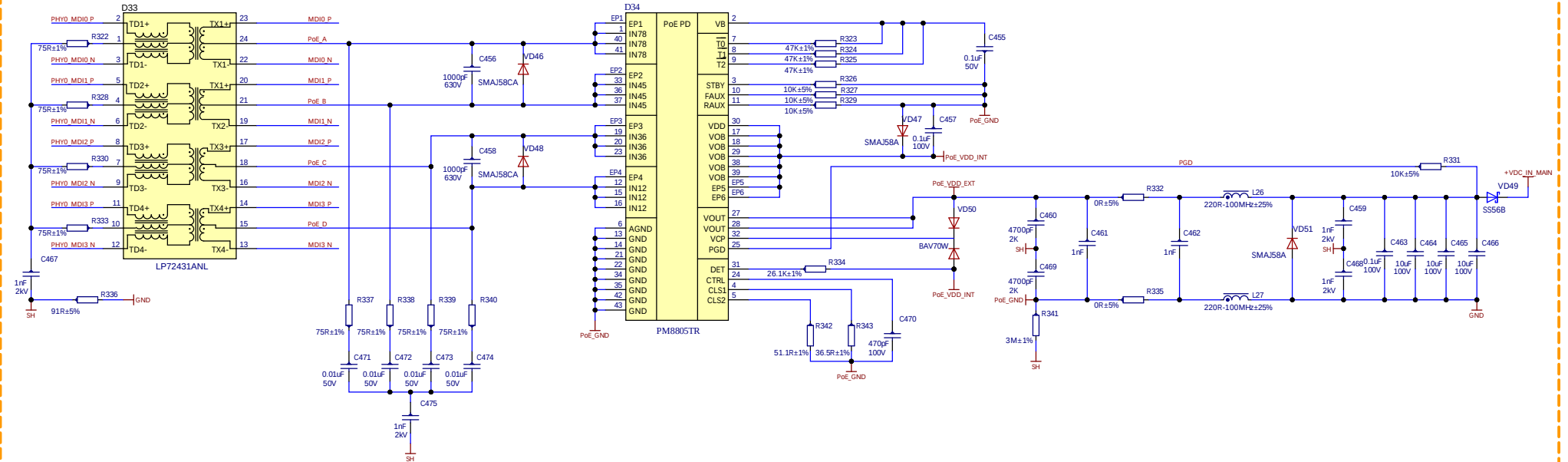
					ЮПТН.466256.00233						
Изм	Лист	№ докум.	Подп.	Дата	Модуль процессорный МУ-RK3576J-MB Схема электрическая принципиальная		Лист	Масса	Масштаб		
								-	-		
							Лист 1	Листов 31			
					Н. контр.				000 «МЮ»		
					Умб.						



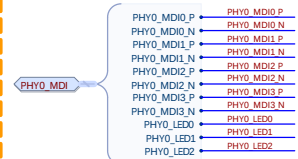
Инб. № подл.	Подп. и дата
Взам. инб. №	Инб. № дубл.

Изм.	Лист	№ докум.	Подп.	Дата

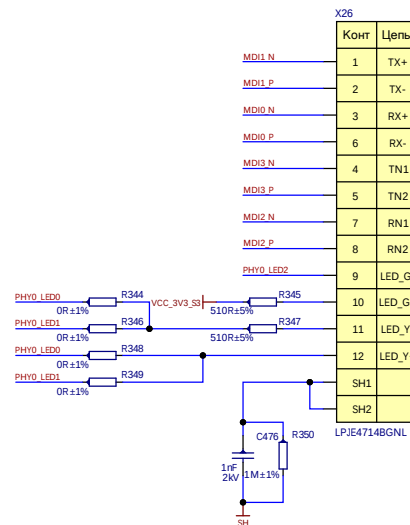
1000BASE-T PoE PD



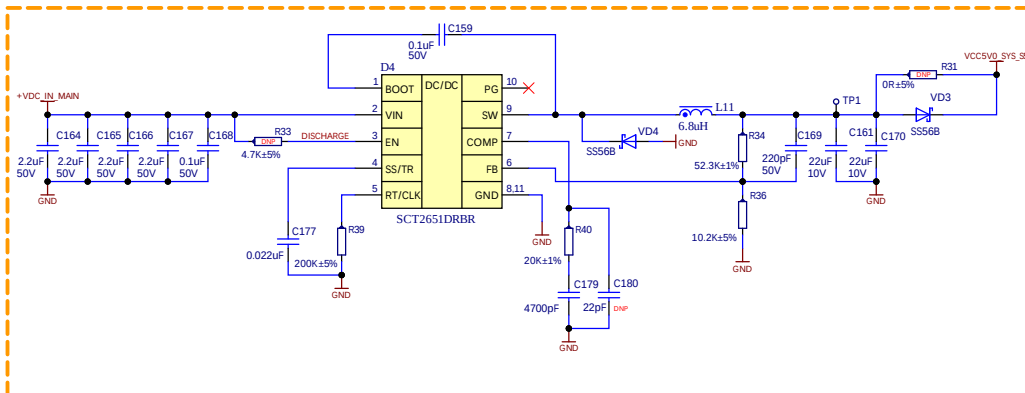
Connection to RGMII0_ETH_PHY



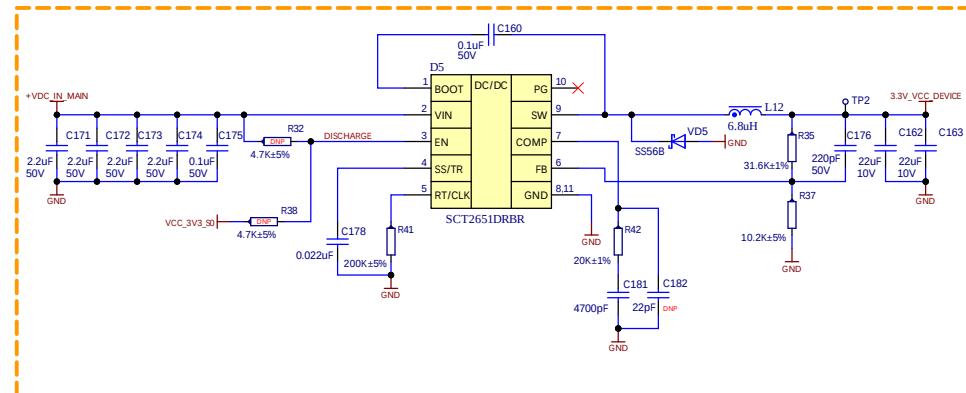
ETH Connector RJ45



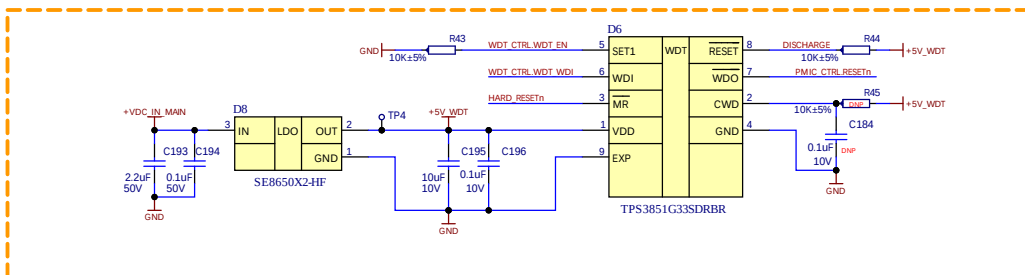
5V_SYS



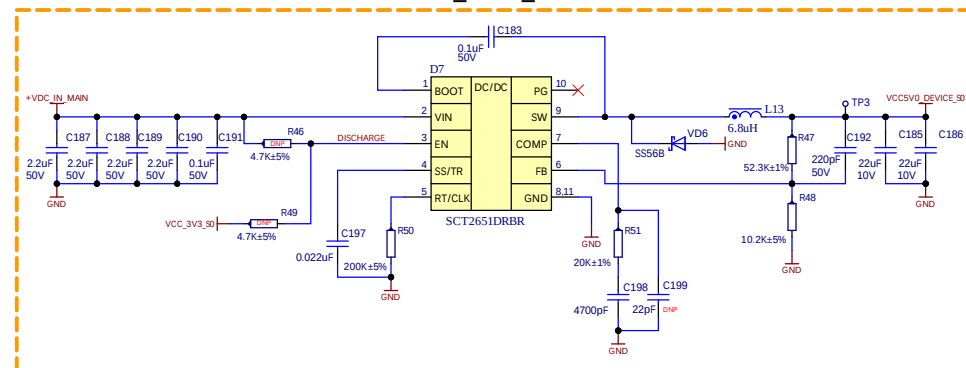
3.3V_VCC_DEVICE



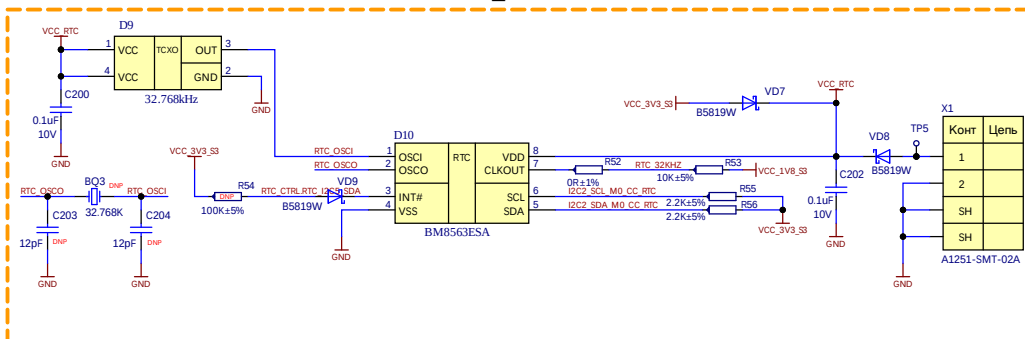
WDT



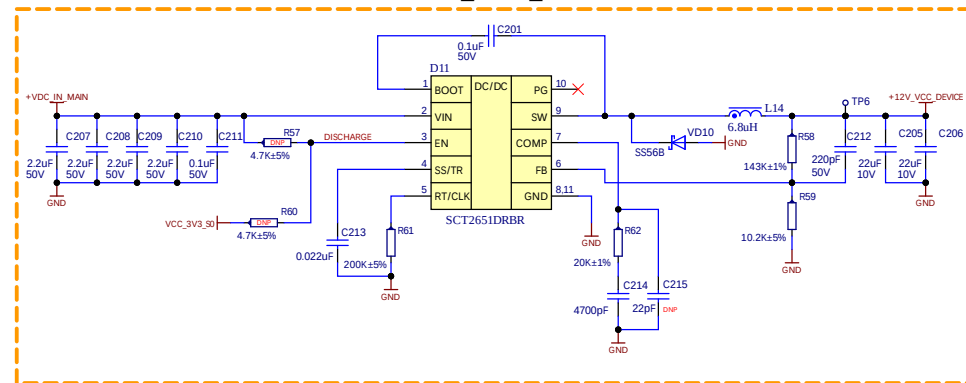
5V_VCC_DEVICE



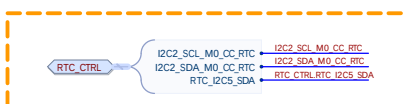
VBAT_RTC



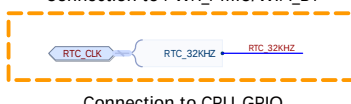
12V_VCC_DEVICE



Connection to CPU_GPIO



Connection to PWR_PMIC/WIFI_BT



Connection to PWR_PMIC/WIFI_BT

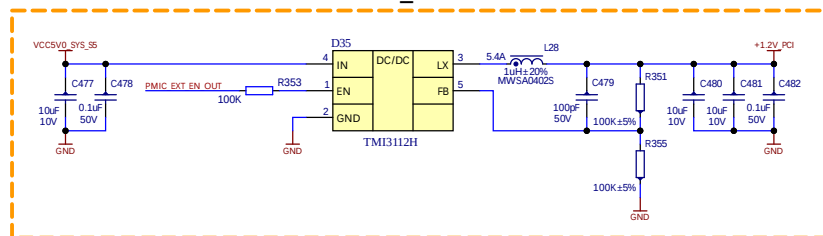


Connection to CPU_GPIO

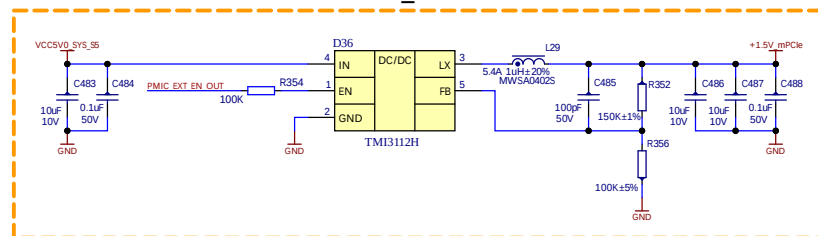


Изм.	Лист	№ докум.	Подп.	Дата

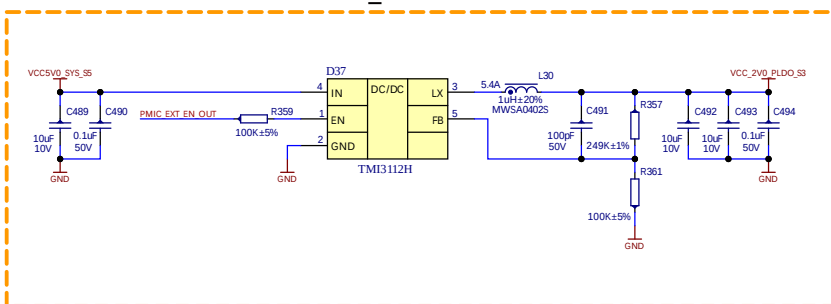
1.2V_PCI



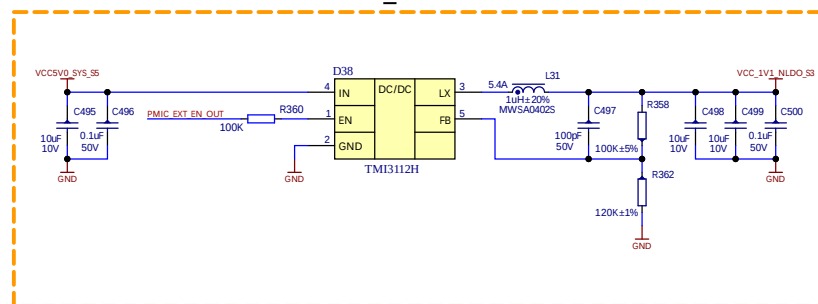
1.5V_PCI



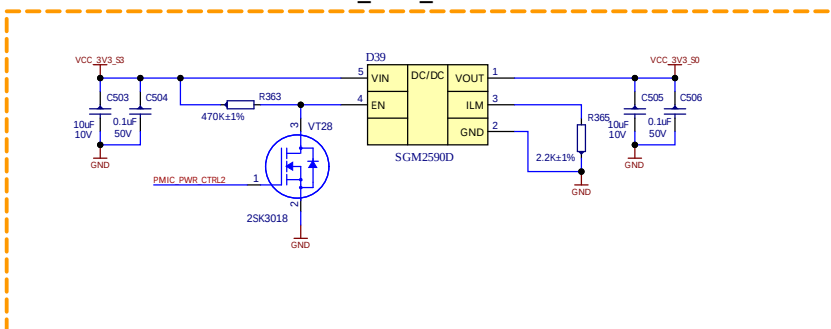
2V_PLDO



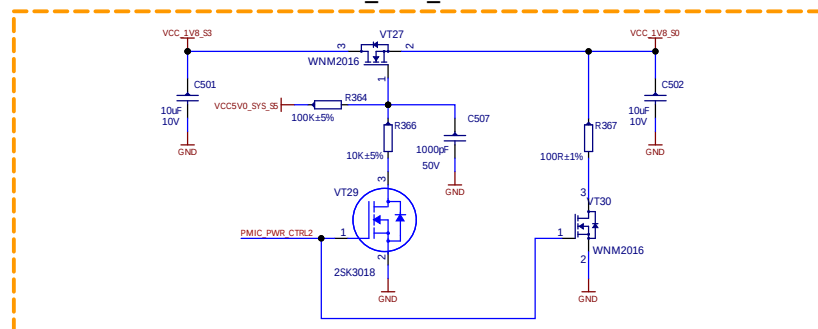
1.1V_NLDO



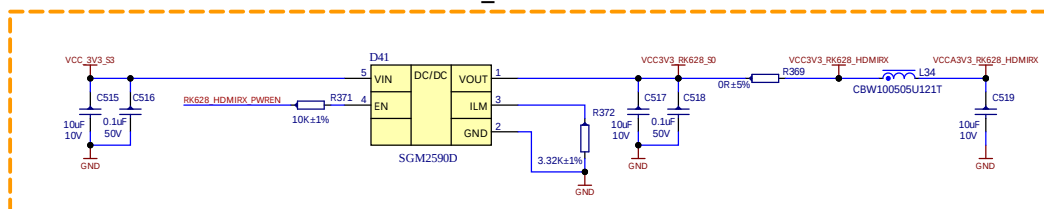
3V3_ILM_EXT



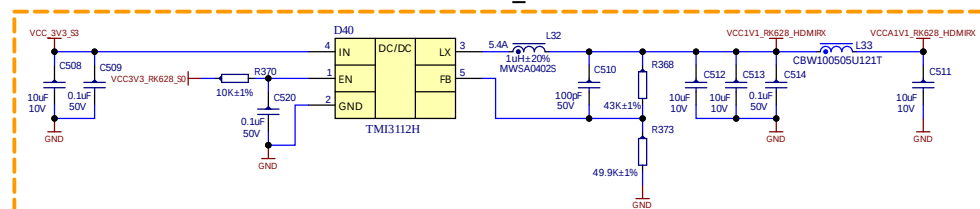
1V8_ILM_EXT



3V3_RK628



1V1_RK628



Connection to CPU_GPIO

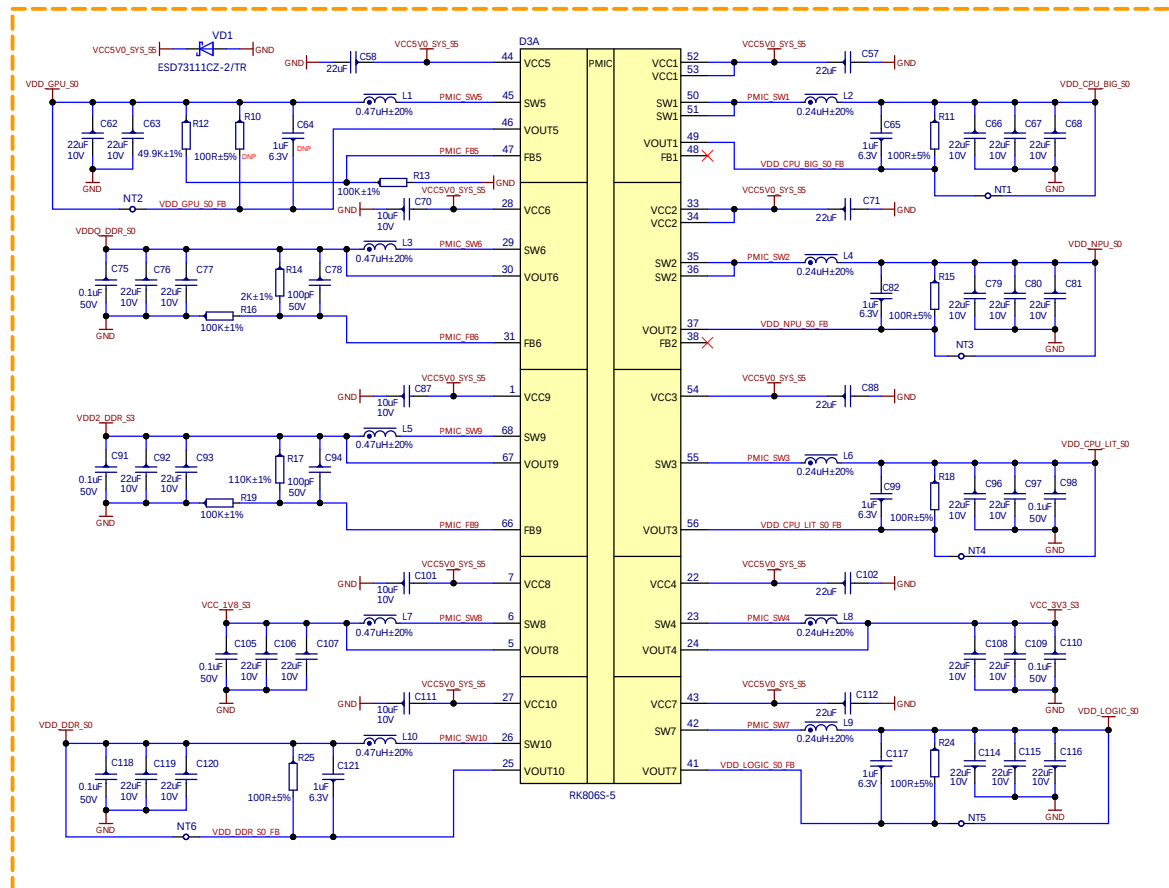
Connection to CPU_GPIO

Connection to CPU_GPIO

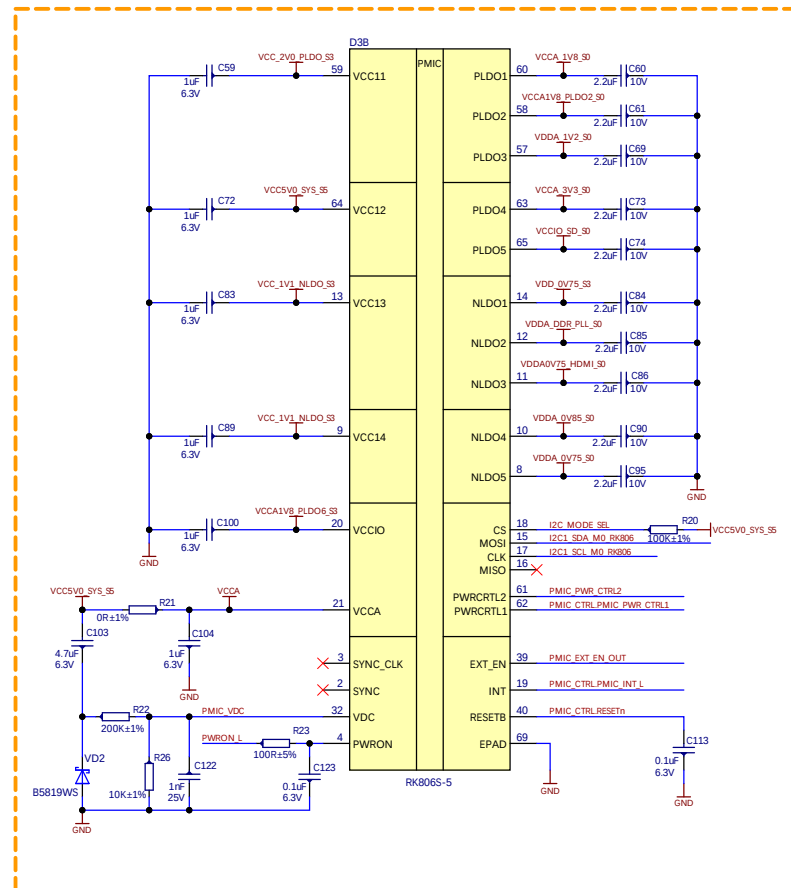


Изм.	Лист	№ докум.	Подп.	Дата

PMIC DC/DC



PMIC LDO/MANAGEMENT



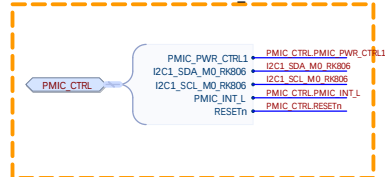
Connection to CPU_GPIO



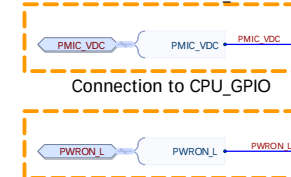
Connection to CPU_GPIO



Connection to CPU_GPIO



Connection to CPU_GPIO

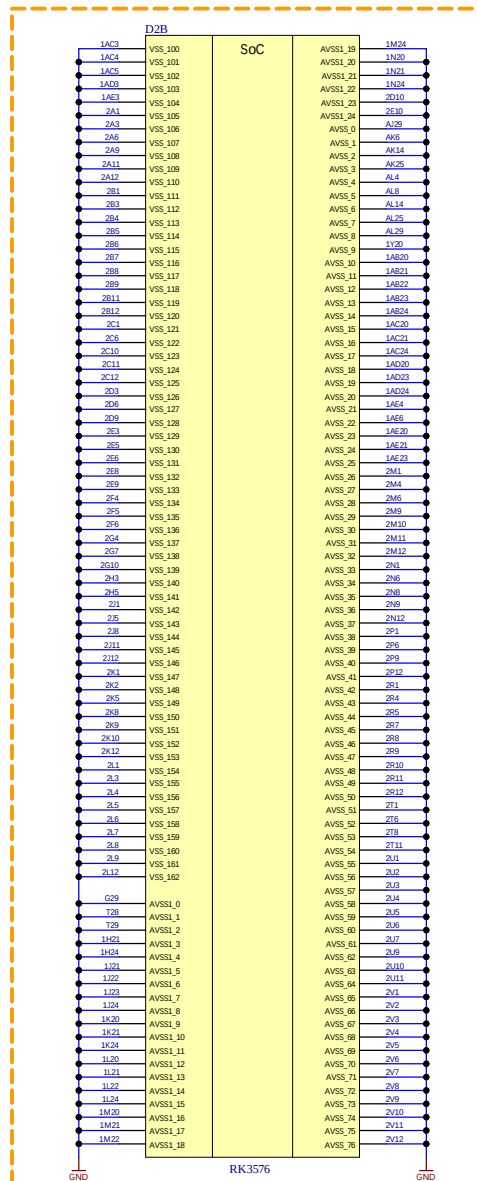


Connection to CPU_GPIO

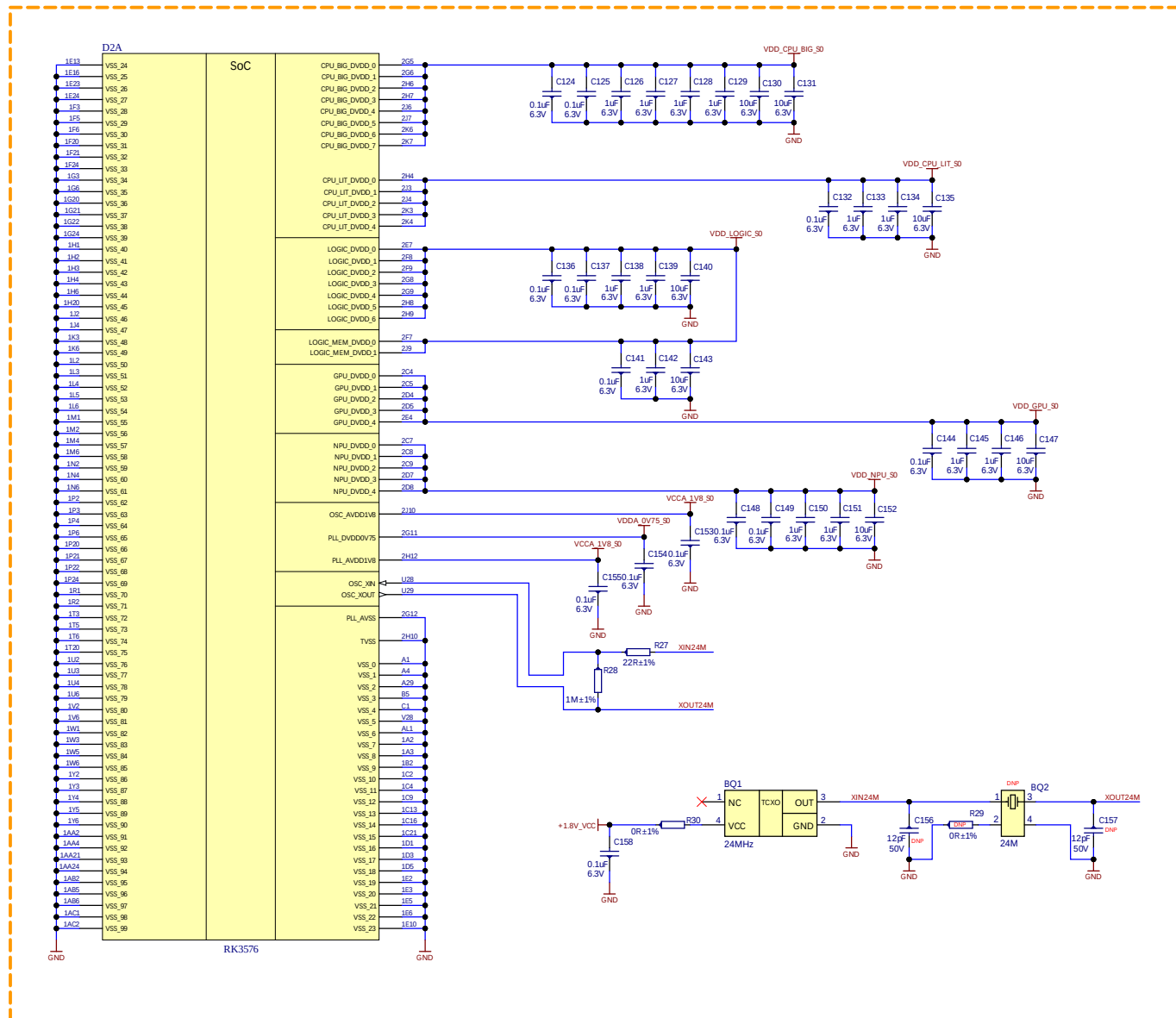


Изм.	Лист	№ докум.	Подп.	Дата

CPU GND



CPU PWR



Подп. и дата

Взам. инб. №

Подп. и дата

Инб. № подл.

Изм.	Лист	№ докум.	Подп.	Дата

ЮПТН.466256.00233

Копировал

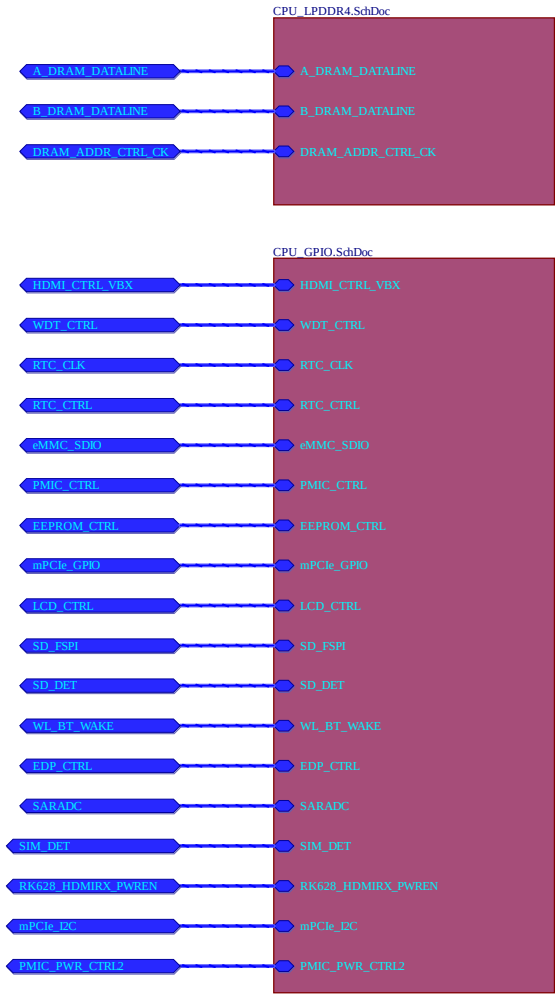
Формат А3

Лист

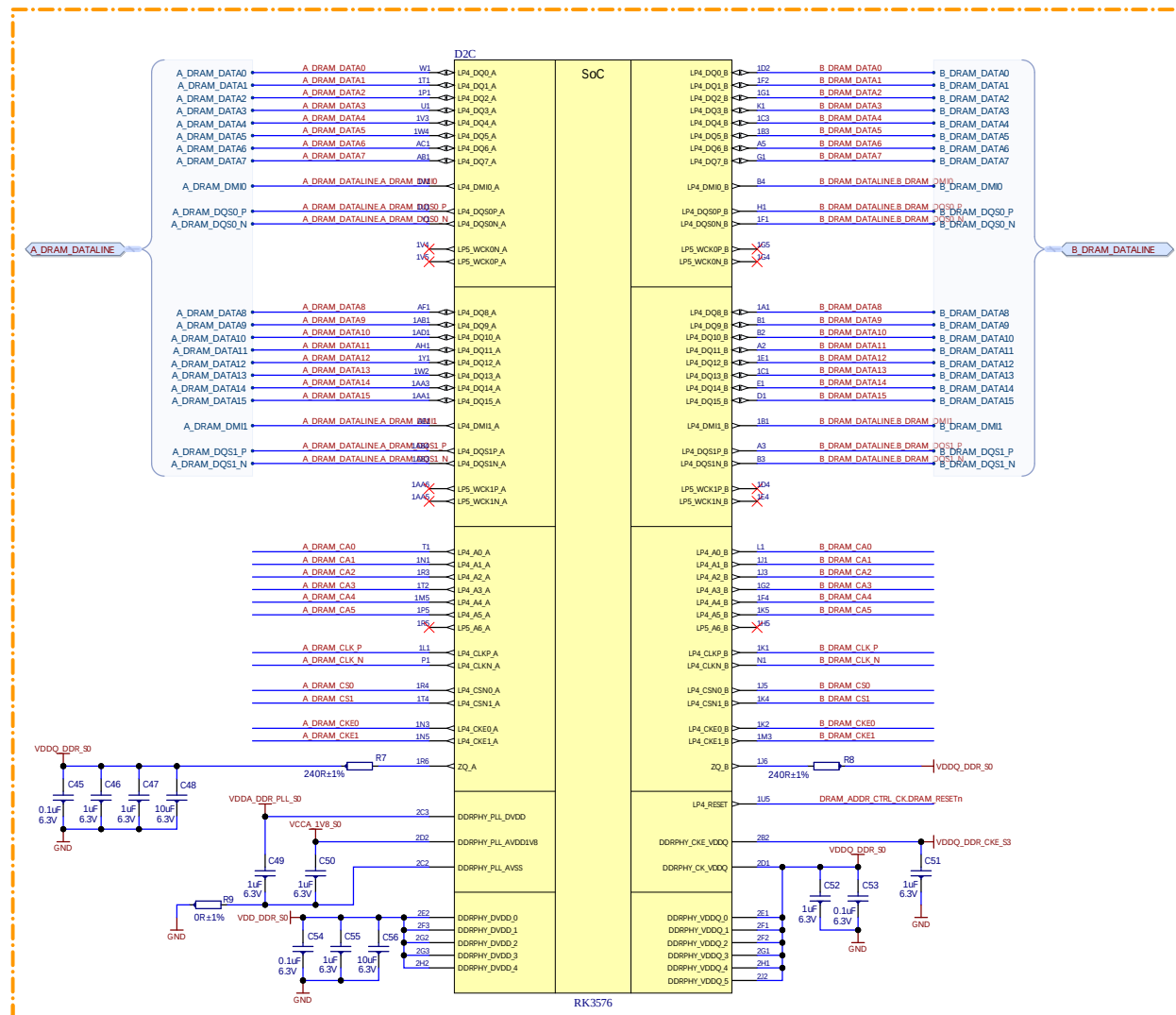
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Инб. № подл.	Подп. и дата	Взам. инб. №	Инб. № дубл.	Подп. и дата

ЮПТН.466256.00233



Блок подключения CPU к LPDDR4



Connection to CPU_GPIO



Подп. и дата

Инв. № з/усл.

Взам. инв. №

Подп. и дата

Инв. № подл.

Изм.	Лист	№ докум.	Подп.	Дата

Connection to PWR_PMIC/WIFI_BT

SD_DET → SDMMC0_DET → SD_DET.SDMMC0_DET

Connection to CPU_GPIO

WL_BT_WAKE → WL_BT_WAKE.WL_BT_WAKE.WL_HOST_WAKE
BT_WAKE → BT_WAKE

Connection to CPU_GPIO

EEPROM_CTRL → EEPROM_WP_CPU_0 → EEPROM_CTRL.EEPROM_WP_CPU_0
EEPROM_CTRL → EEPROM_WP_CPU_1 → EEPROM_CTRL.EEPROM_WP_CPU_1
EEPROM_CTRL → I2C6_SCL_M3 → EEPROM_CTRL.I2C6_SCL_M3
EEPROM_CTRL → I2C6_SDA_M3 → EEPROM_CTRL.I2C6_SDA_M3

Connection to PWR_PMIC/WIFI_BT

PMIC_CTRL → PMIC_CTRL.PMIC_CTRL.PMIC_CTRL1
I2C1_SDA_M0_RK806 → I2C1_SDA_M0_RK806
I2C1_SCL_M0_RK806 → I2C1_SCL_M0_RK806
PMIC_INT_L → PMIC_CTRL.PMIC_INT_L
RESETn → PMIC_CTRL.RESETn

Connection to PWR_PMIC/WIFI_BT

RTC_CLK → RTC_32KHZ → RTC_32KHZ

Connection to V-by-One

HDMI_CTRL_VBX → HDMI_TX_SCL → HDMI_CTRL_VBX.HDMI_TX_SCL
HDMI_CTRL_VBX → HDMI_TX_SDA → HDMI_CTRL_VBX.HDMI_TX_SDA
HDMI_CTRL_VBX → HDMI_TX_CEC → HDMI_CTRL_VBX.HDMI_TX_CEC
HDMI_CTRL_VBX → HDMI_TX_HPDIN → HDMI_CTRL_VBX.HDMI_TX_HPDIN

Connection to CPU_GPIO

SD_FSP1 → SD_FSP1.D0 → SD_FSP1.D0
SD_FSP1 → SD_FSP1.D1 → SD_FSP1.D1
SD_FSP1 → SD_FSP1.D2 → SD_FSP1.D2
SD_FSP1 → SD_FSP1.D3 → SD_FSP1.D3
SD_FSP1 → SD_FSP1.CMD → SD_FSP1.CMD
SD_FSP1 → SD_FSP1.CSN → SD_FSP1.CSN
SD_FSP1 → SD_FSP1.CLK → SD_FSP1.CLK

Connection to CPU_GPIO

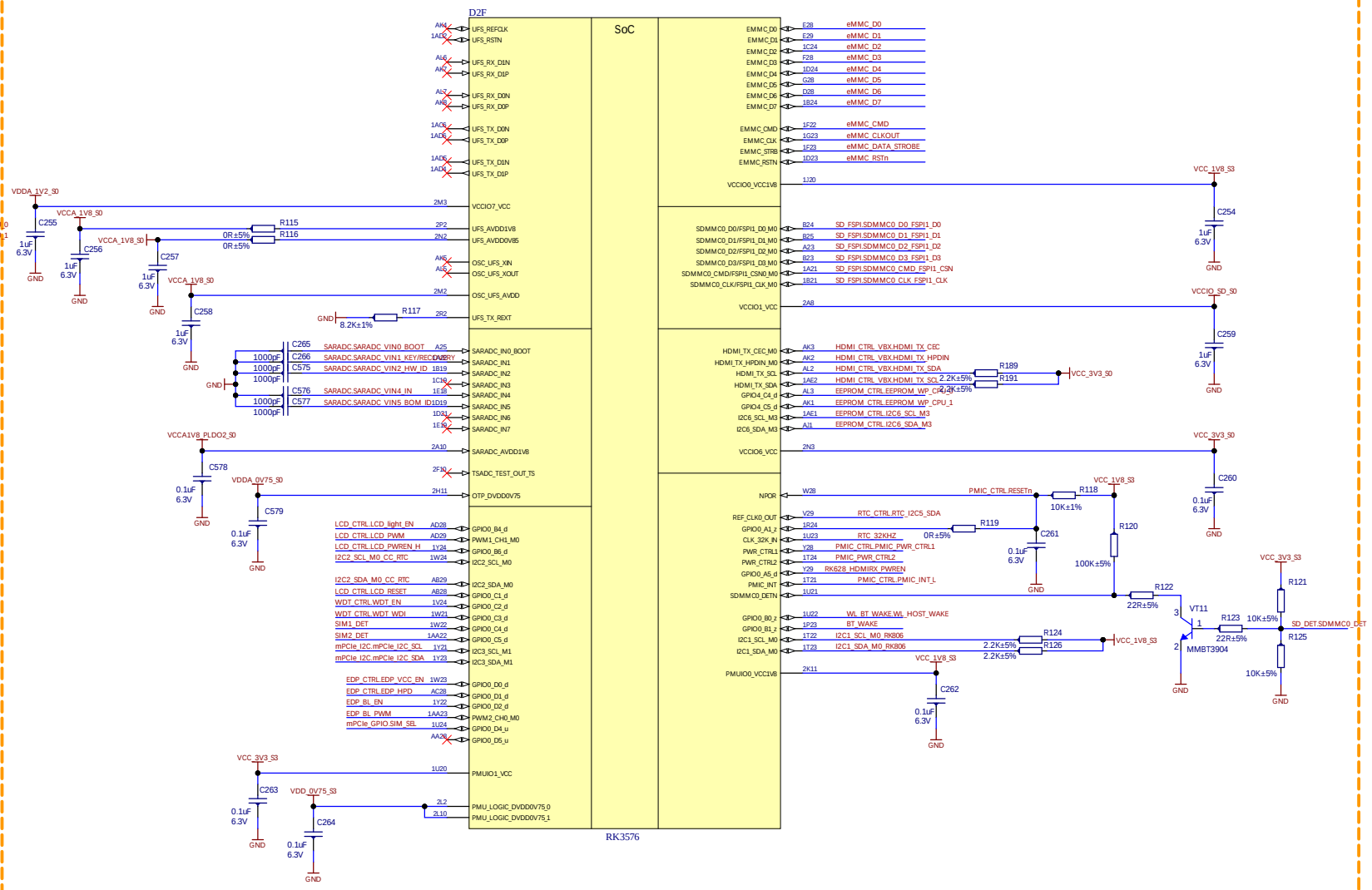
eMMC_SDIO → eMMC_D0 → eMMC_D0
eMMC_SDIO → eMMC_D1 → eMMC_D1
eMMC_SDIO → eMMC_D2 → eMMC_D2
eMMC_SDIO → eMMC_D3 → eMMC_D3
eMMC_SDIO → eMMC_D4 → eMMC_D4
eMMC_SDIO → eMMC_D5 → eMMC_D5
eMMC_SDIO → eMMC_D6 → eMMC_D6
eMMC_SDIO → eMMC_D7 → eMMC_D7
eMMC_CMD → eMMC_CMD
eMMC_CLKOUT → eMMC_CLKOUT
eMMC_DATA_STROBE → eMMC_DATA_STROBE
eMMC_RSTn → eMMC_RSTn

Connection to CPU_GPIO

RTC_CTRL → I2C2_SCL_M0_CC_RTC → I2C2_SCL_M0_CC_RTC
RTC_CTRL → I2C2_SDA_M0_CC_RTC → I2C2_SDA_M0_CC_RTC
RTC_CTRL → I2C5_SCL → I2C5_SCL
RTC_CTRL → I2C5_SDA → I2C5_SDA

Connection to PWR_MAIN

WDT_CTRL → WDT_EN → WDT_CTRL.WDT_EN
WDT_CTRL → WDT_WDI → WDT_CTRL.WDT_WDI
RESETn → PMIC_CTRL.RESETn



Connection to Peripheral

SARADC → SARADC_VIN0_BOOT → SARADC.SARADC_VIN0_BOOT
SARADC → SARADC_VIN1_KEYRECOVERY → SARADC.SARADC_VIN1_KEYRECOVERY
SARADC → SARADC_VIN2_HW_ID → SARADC.SARADC_VIN2_HW_ID
SARADC → SARADC_VIN4_IN → SARADC.SARADC_VIN4_IN
SARADC → SARADC_VIN5_BOM_ID → SARADC.SARADC_VIN5_BOM_ID

Connection to MIPI DSI CSI

LCD_CTRL → LCD_LIGHT_EN → LCD_CTRL.LCD_LIGHT_EN
LCD_CTRL → LCD_PWM → LCD_CTRL.LCD_PWM
LCD_CTRL → LCD_PWREN_H → LCD_CTRL.LCD_PWREN_H
LCD_CTRL → LCD_RESET → LCD_CTRL.LCD_RESET

Connection to LVDS eDP

EDP_CTRL → EDP_VCC_EN → EDP_CTRL.EDP_VCC_EN
EDP_CTRL → EDP_BL_EN → EDP_CTRL.EDP_BL_EN
EDP_CTRL → EDP_BL_PWM → EDP_CTRL.EDP_BL_PWM
EDP_CTRL → EDP_HPD → EDP_CTRL.EDP_HPD

Connection to CPU_GPIO

RK628_HDMIRX_PWREN → RK628_HDMIRX_PWREN → RK628_HDMIRX_PWREN

Connection to CPU_GPIO

PMIC_PWR_CTRL2 → PMIC_PWR_CTRL2 → PMIC_PWR_CTRL2

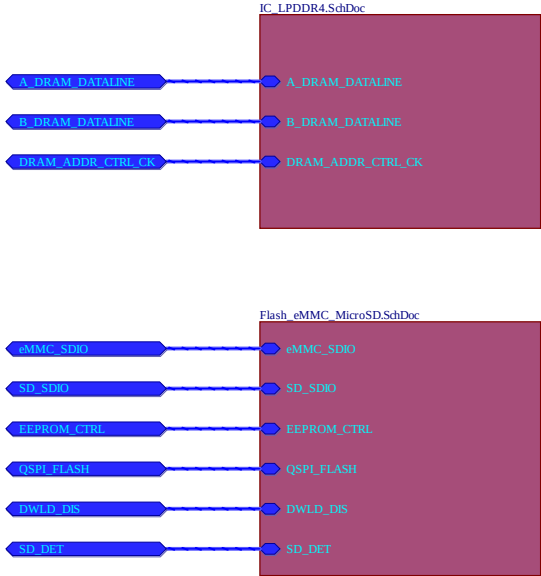
Connection to CPU_GPIO

SIM_DET → SIM1_DET → SIM1_DET
SIM_DET → SIM2_DET → SIM2_DET
mPCIe_I2C → mPCIe_I2C_SCL → mPCIe_I2C.mPCIe_I2C_SCL
mPCIe_I2C → mPCIe_I2C_SDA → mPCIe_I2C.mPCIe_I2C_SDA
mPCIe_GPIO → SIM_SEL → mPCIe_GPIO.SIM_SEL

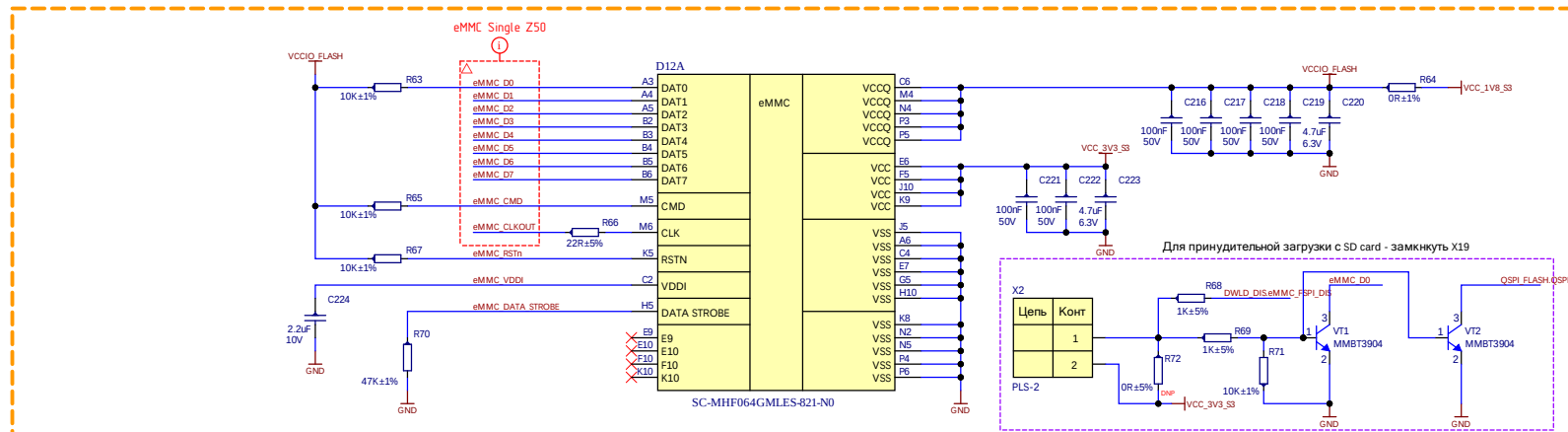
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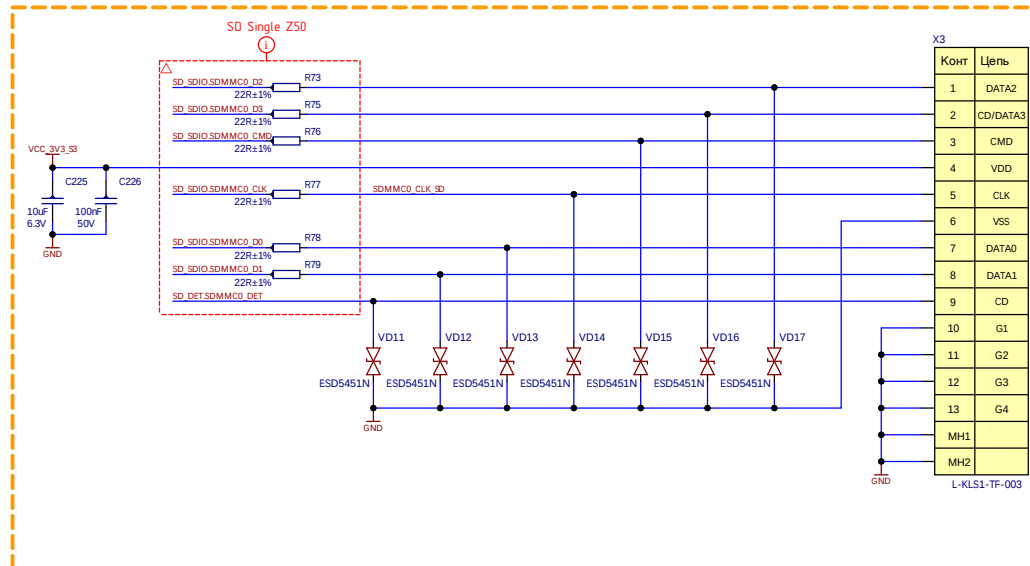
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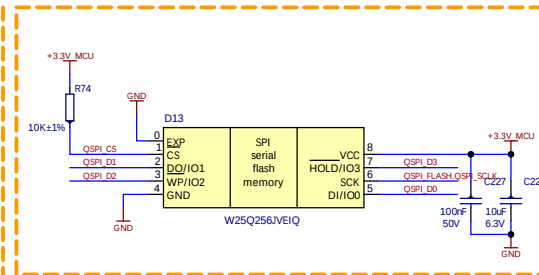
eMMC



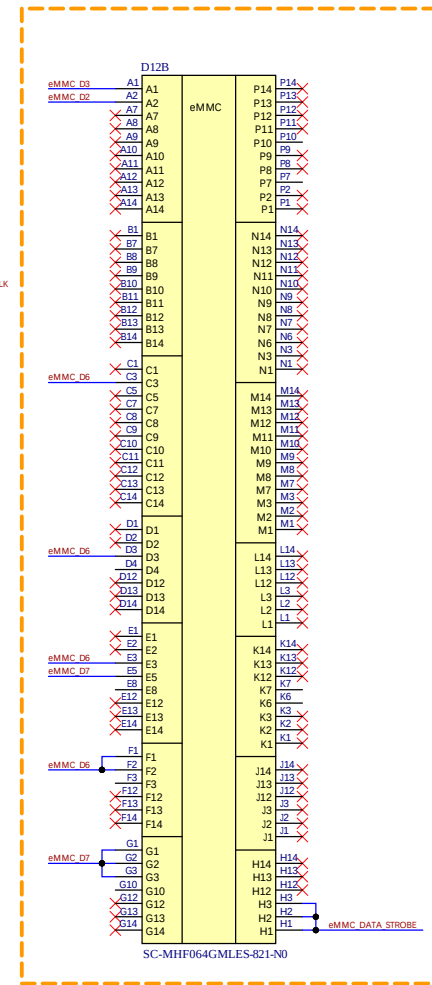
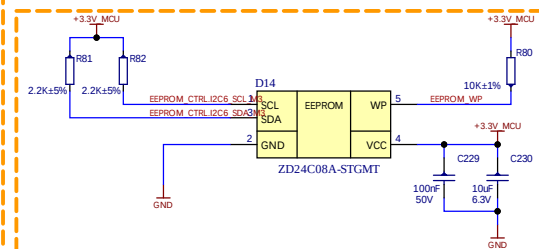
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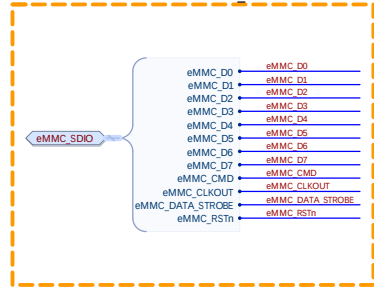
Flash



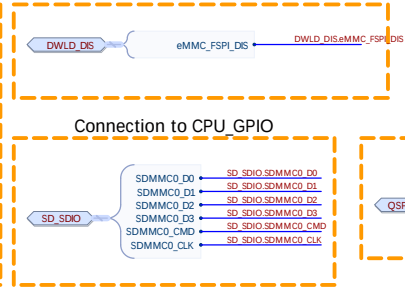
EEPROM



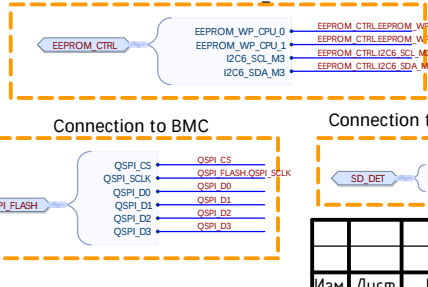
Connection to CPU_GPIO



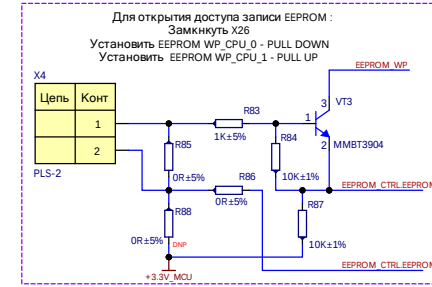
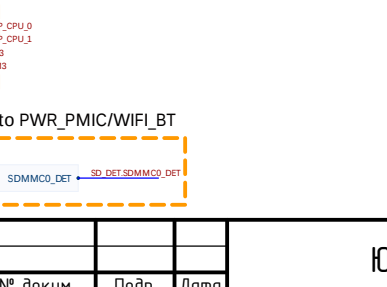
Connection to BMC



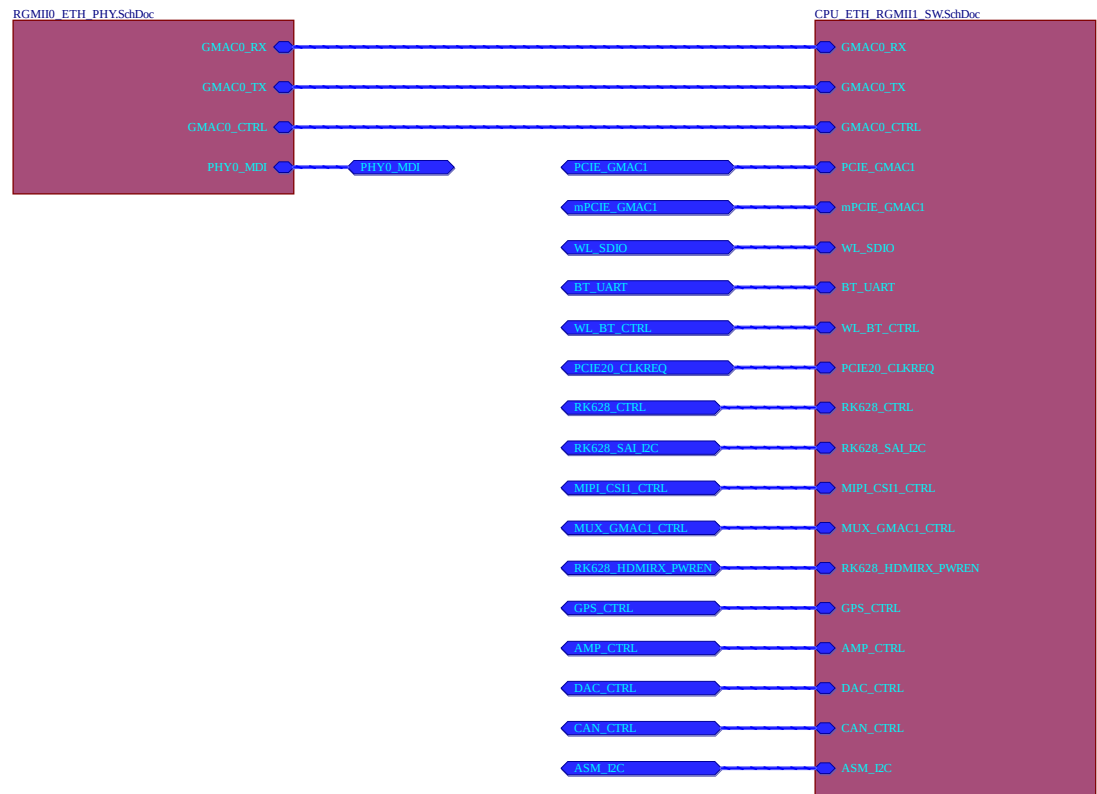
Connection to CPU_GPIO



Connection to PWR_PMIC/WIFI_BT



Инб. № подл.	Подп. и дата
Взам. инб. №	Инб. № дубл.



A diagram showing a set of pins labeled PHY0_MDI0_P through PHY0_LED2. A bracket on the left groups the first six pins (PHY0_MDI0_P to PHY0_MDI3_N) under the label PHY0_MDI. The pins are connected to a vertical line on the right, which then branches out to individual labels. The connections are as follows:

Pin Label	Destination Label
PHY0_MDI0_P	PHY0_MDI0_P
PHY0_MDI0_N	PHY0_MDI0_N
PHY0_MDI1_P	PHY0_MDI1_P
PHY0_MDI1_N	PHY0_MDI1_N
PHY0_MDI2_P	PHY0_MDI2_P
PHY0_MDI2_N	PHY0_MDI2_N
PHY0_MDI3_P	PHY0_MDI3_P
PHY0_MDI3_N	PHY0_MDI3_N
PHY0_LED0	PHY0_LED0
PHY0_LED1	PHY0_LED1
PHY0_LED2	PHY0_LED2

Figure 1: MAC/PHY Register Map

Register Name	Address Range
GMACD_RXD0	0x00000000 - 0x00000007
GMACD_RXD1	0x00000008 - 0x0000000F
GMACD_RXD2	0x00000010 - 0x0000001F
GMACD_RXD3	0x00000020 - 0x0000002F
GMACD_RXD4	0x00000030 - 0x0000003F
GMACD_RXD5	0x00000040 - 0x0000004F
GMACD_RXD6	0x00000050 - 0x0000005F
GMACD_RXD7_CRS	0x00000060 - 0x0000006F
GMACD_RXD7_CRS2	0x00000070 - 0x0000007F
GMACD_TXD0	0x00000080 - 0x00000087
GMACD_TXD1	0x00000088 - 0x0000008F
GMACD_TXD2	0x00000090 - 0x0000009F
GMACD_TXD3	0x000000A0 - 0x000000AF
GMACD_TXD4	0x000000B0 - 0x000000BF
GMACD_TXD5	0x000000C0 - 0x000000CF
GMACD_TXD6	0x000000D0 - 0x000000DF
GMACD_TXD7	0x000000E0 - 0x000000EF
GMACD_TXD8	0x000000F0 - 0x000000FF
PHY0_RSTN	0x00000100 - 0x00000107
PHY0_MDC	0x00000108 - 0x0000010F
PHY0_MDIO	0x00000110 - 0x0000011F
PHY0_CLKOUT	0x00000120 - 0x0000012F

PHY0 Address Config (ADDR_01)

4.7kΩ±1% R295 PHY0_PHYA00 4.7kΩ±1% R296 VCCIO_PHY0

4.7kΩ±1% R298 PHY0_PHYA01 4.7kΩ±1% R299

4.7kΩ±1% R301 PHY0_PHYA02

GND

Pull-up for additional 2ns delay to RXC/TXC Default On

PHY0 Address Config (ADDR_02)

4.7kΩ±1% R305 PHY0_RXD0 4.7kΩ±1% R306 VCCIO_PHY0

4.7kΩ±1% R307 PHY0_RXD1 4.7kΩ±1% R308

GND

EN PLL @ ALDPS (low power mode) Default - OFF

PHY0 Address Config (ADDR_03)

4.7kΩ±1% R312 PHY0_RXD2 4.7kΩ±1% R313 VCCIO_PHY0

GND

PHY0_IO Voltage Config

Default - Internal 1.8V

4.7kΩ±1% R317 PHY0_LED0 4.7kΩ±1% R318 VCC3V3_PHY0

4.7kΩ±1% R319 PHY0_LED2 4.7kΩ±1% R320

4.7kΩ±1% R321 PHY0_LED1

GND

Ближе к PHY0

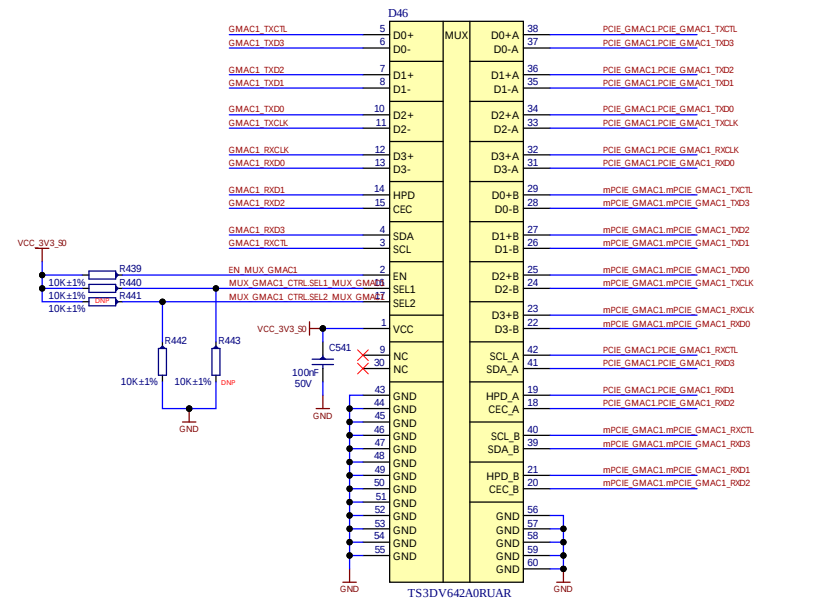
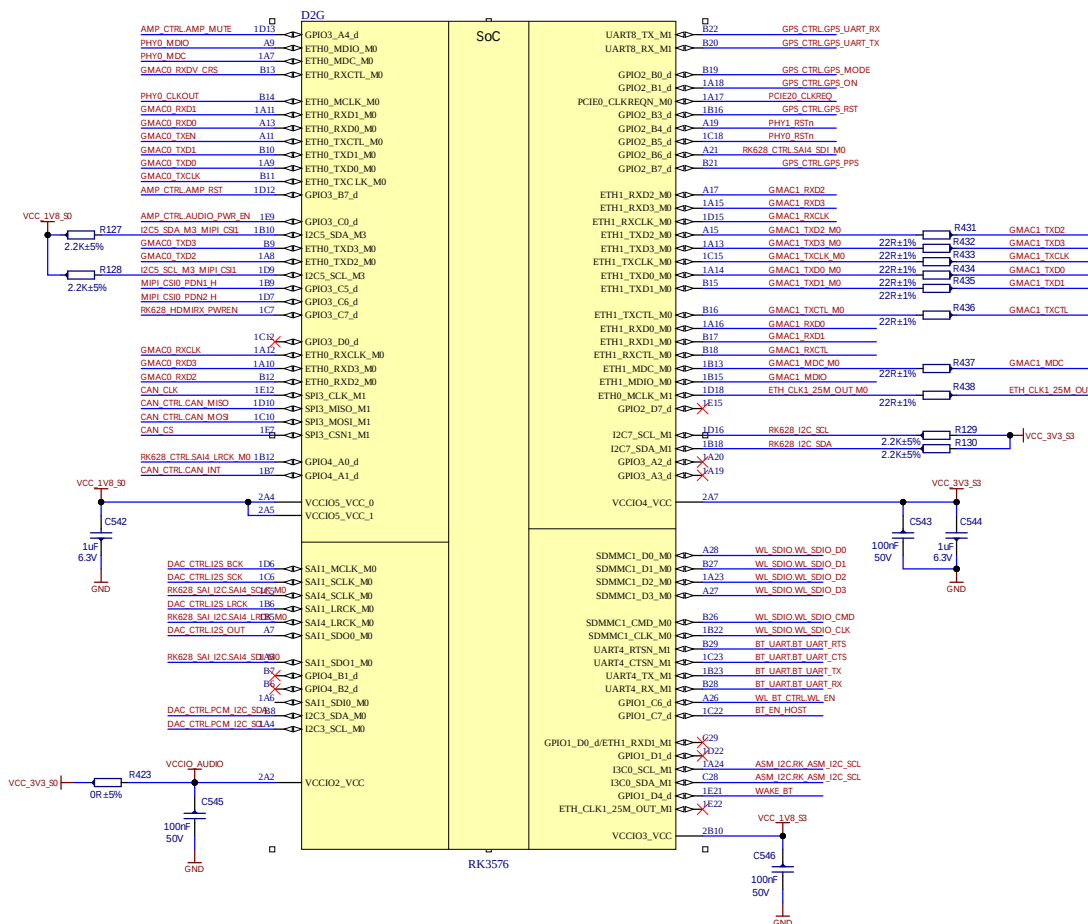
PHY0 Signal	Resistor	Register	GMAC0 Signal
PHY0_RXD0	22R±1%	R294	GMAC0_RXD0
PHY0_RXD1	22R±1%	R297	GMAC0_RXD1
PHY0_RXD2	22R±1%	R300	GMAC0_RXD2
PHY0_PHYAD0	22R±1%	R302	GMAC0_RXD3
PHY0_PHYAD1	22R±1%	R303	GMAC0_RXCLK
PHY0_PHYAD2	22R±1%	R304	GMAC0_RXDV_CRS

Ближе к CPU

PHY0 Signal	Resistor	Register	GMAC0 Signal
PHY0_TXD0	22R±1%	R309	GMAC0_TXD0
PHY0_TXD1	22R±1%	R310	GMAC0_TXD1
PHY0_TXD2	22R±1%	R311	GMAC0_TXD2
PHY0_TXD3	22R±1%	R314	GMAC0_TXD3
PHY0_TXEN	22R±1%	R315	GMAC0_TXEN
PHY0_TXCLK	22R±1%	R316	GMAC0_TXCLK

CPU_ETH

ETH_MUX



Подп. и дата

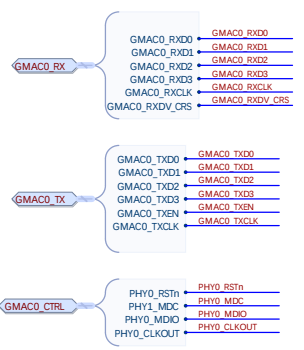
Инд. № докум.

Взам. инд. №

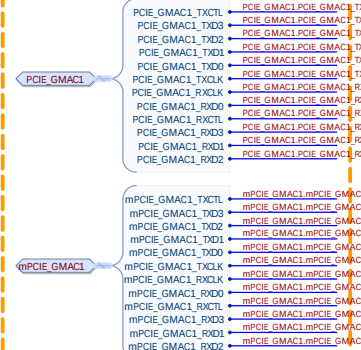
Подп. и дата

Инд. № подл.

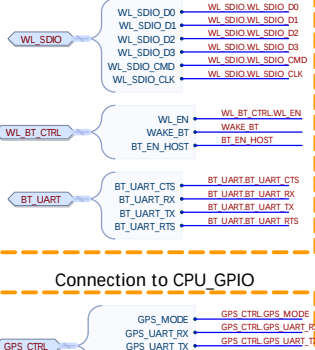
Connection to RGMII0 ETH PHY



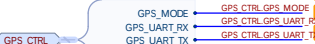
Connection to CPU GPIO



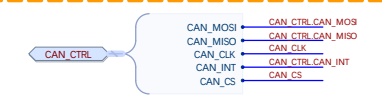
Connection to CPU GPIO



Connection to CPU GPIO



Connection to CAN-FD



Connection to CPU_ETH RGMII1 SW



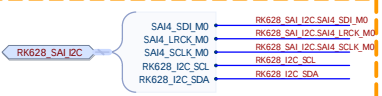
Connection to AUDIO



Connection to CPU_ETH RGMII1 SW



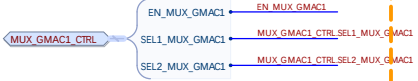
Connection to V-by-One



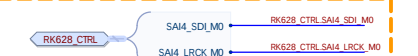
Connection to MIPI DSI CSI



Connection to V-by-One



Connection to CPU_ETH RGMII1 SW



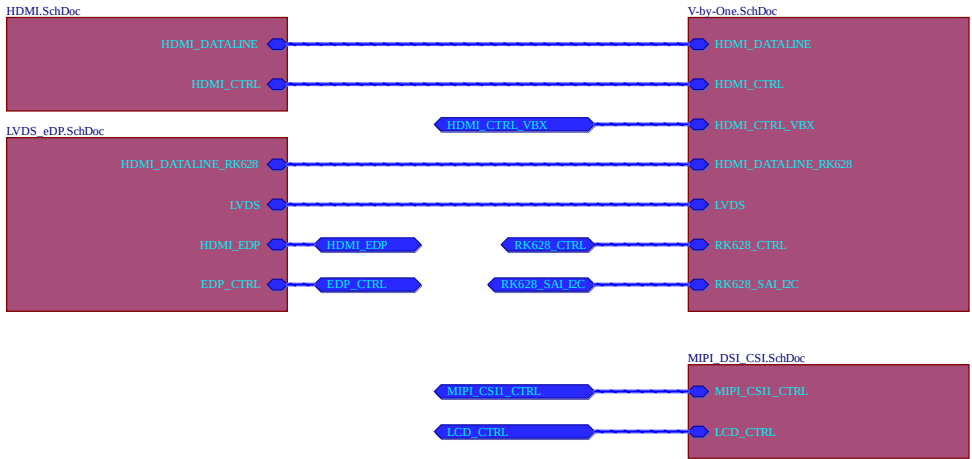
Connection to CPU GPIO



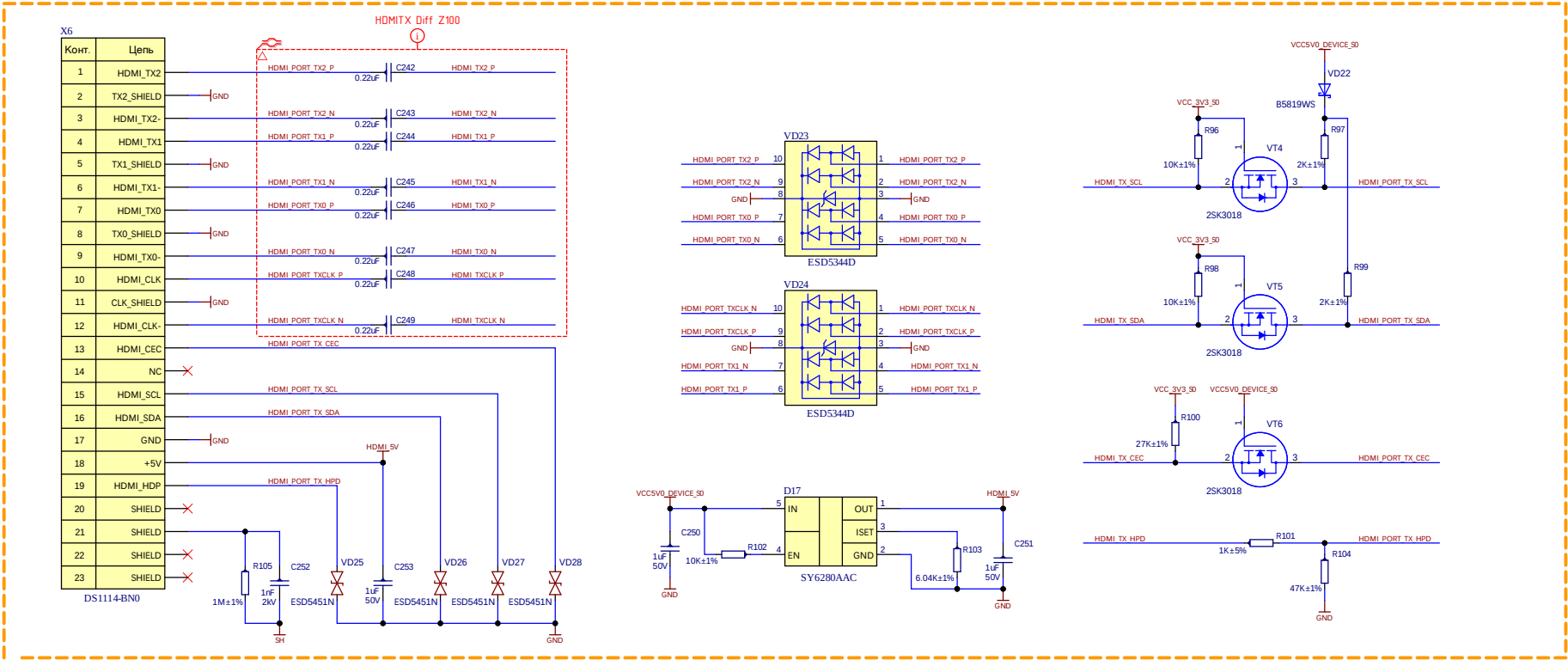
Изм.	Лист	№ докум.	Подп.	Дата
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Инб. № подл.	Подп. и дата	Взам. инб. №	Инб. № дубл.	Подп. и дата

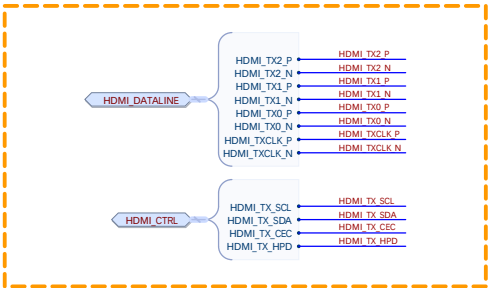
ЮРТИ.466256.00233



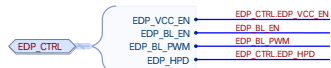
HDMI



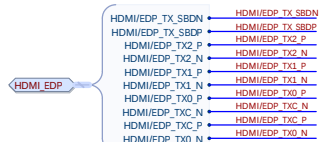
Connection to V-by-One



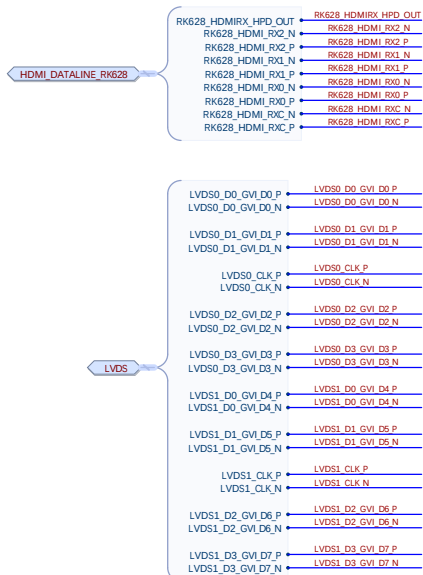
Connection to CPU_GPIO



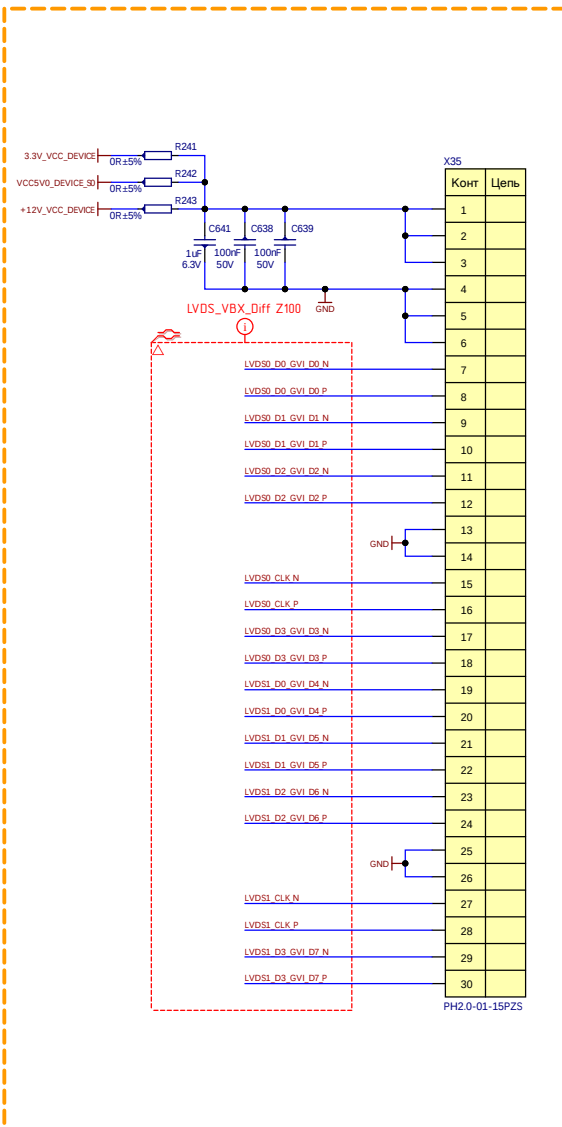
Connection to CPU_PCIE_USB_HDMI_EXT_PCIEX2_mPCIeX2



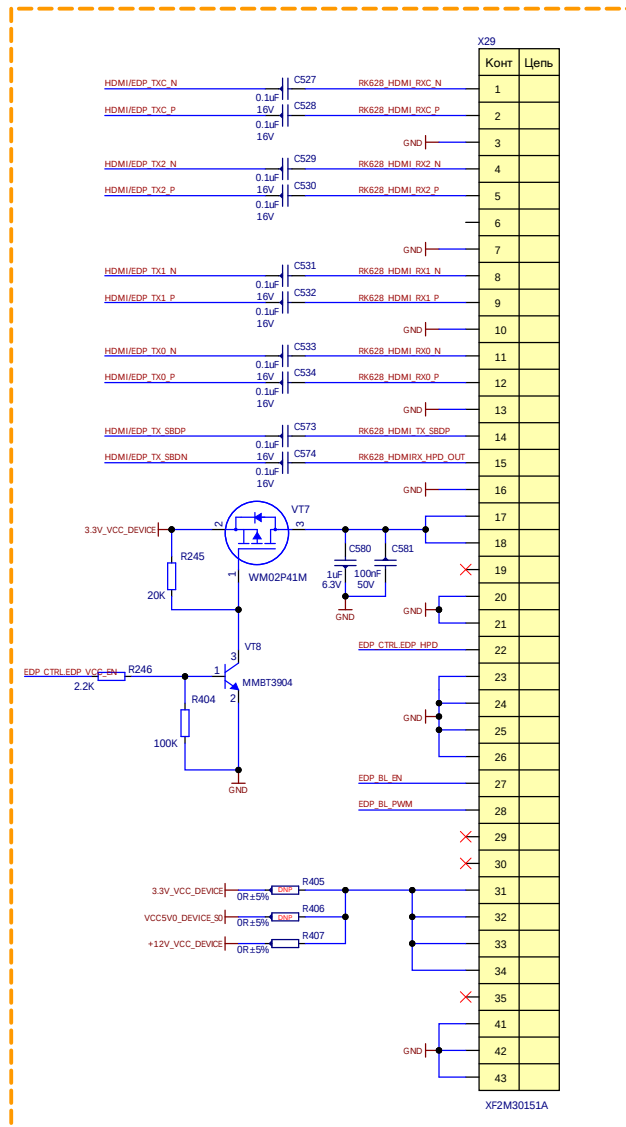
Connection to V-by-One



LVDS



eDP



Подн. и дата

Инд. № дубл.

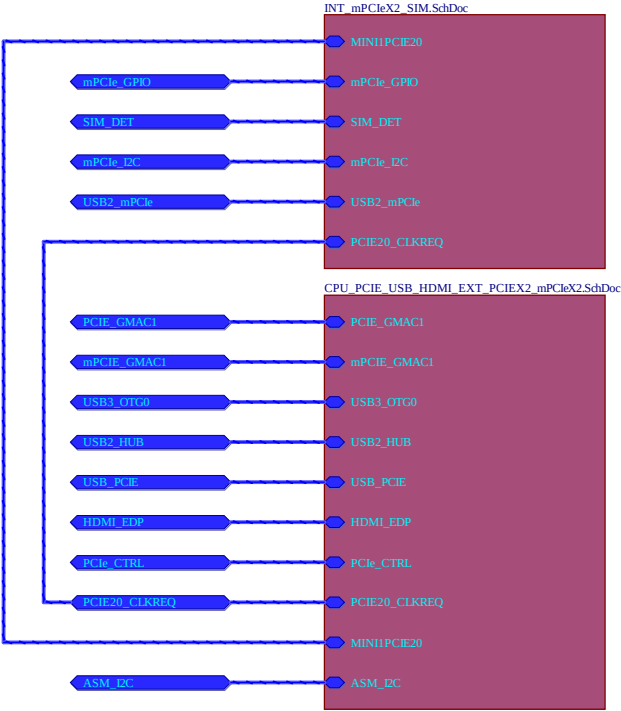
Взам. инд. №

Подн. и дата

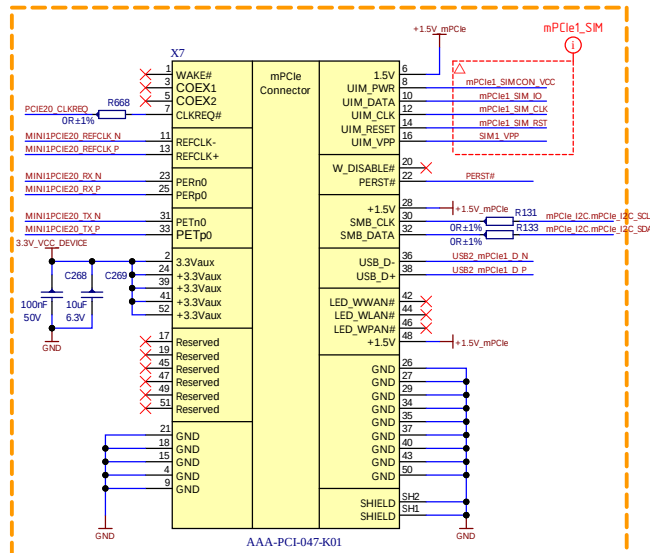
Инд. № подл.

Инб. № подл.	Подп. и дата	Взам. инб. №	Инб. № дубл.	Подп. и дата

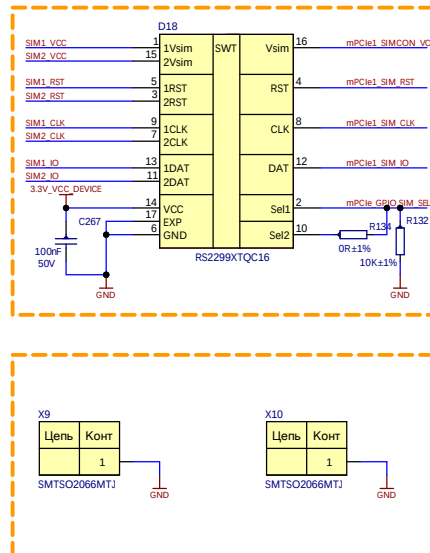
ЮРTH.466256.00233



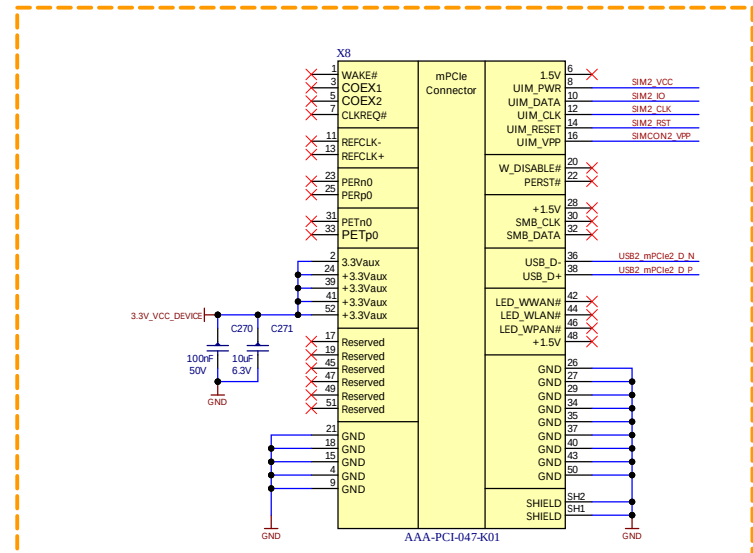
mPCIe1 FULL



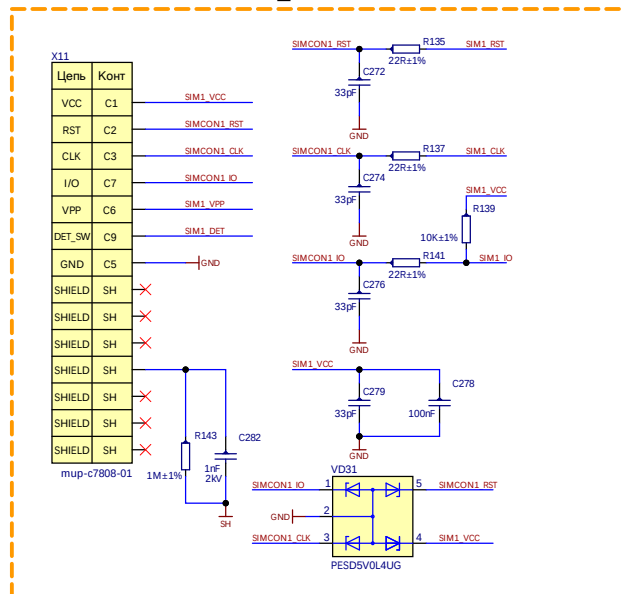
SIM_MUX



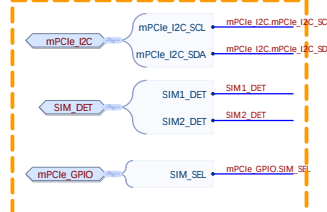
mPCIe2 Only USB



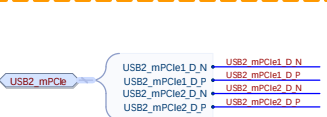
SIM1_Connector



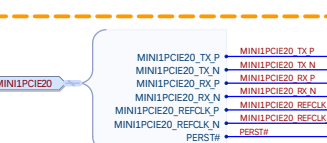
Connection to CPU GPIO



Connection to USB



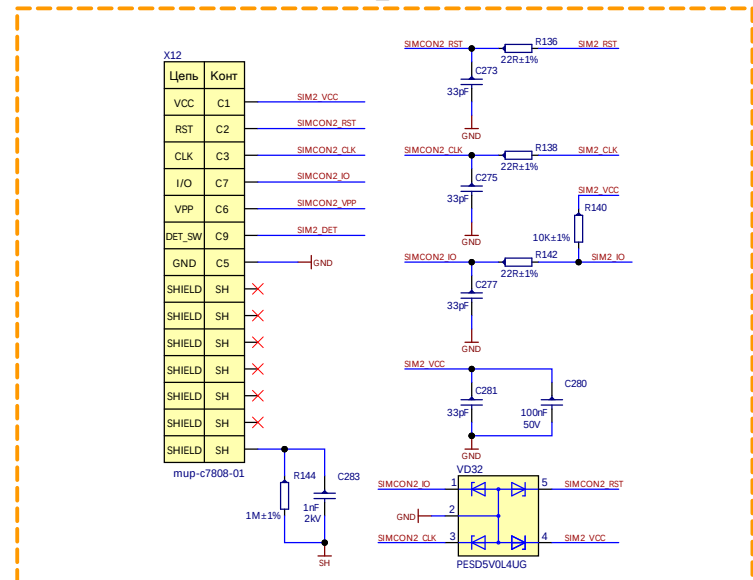
Connection to BMC



Connection to CPU_ETH_RGMII1_SW



SIM2_Connector



Подн. и дата

Взам. инв. №

Подн. и дата

Инд. № подл.

Изм.	Лист	№ докум.	Подн.	Дата
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ИПТН.466256.00233

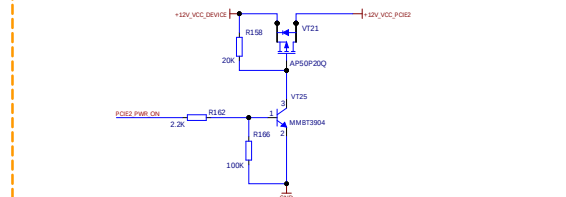
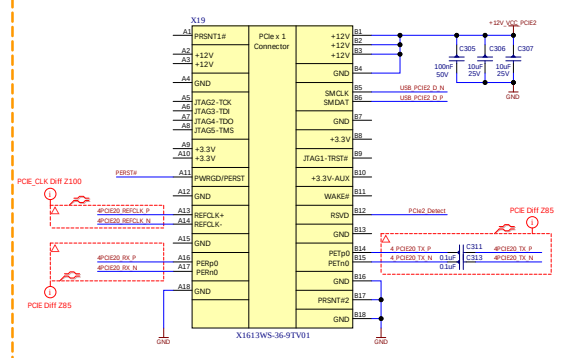
Конструктор

Формат А3

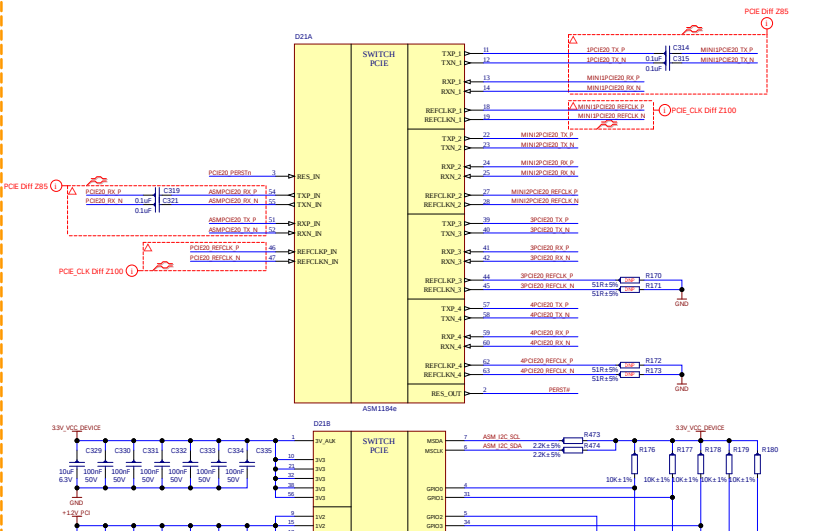
Лист

24

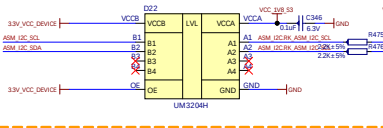
Proprietary PCIE2



PCIE SWITCH

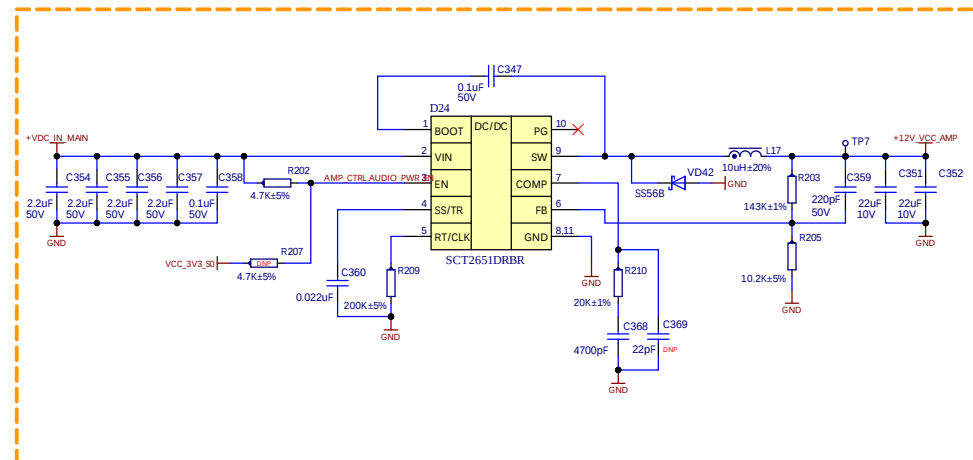


Connection to BMC

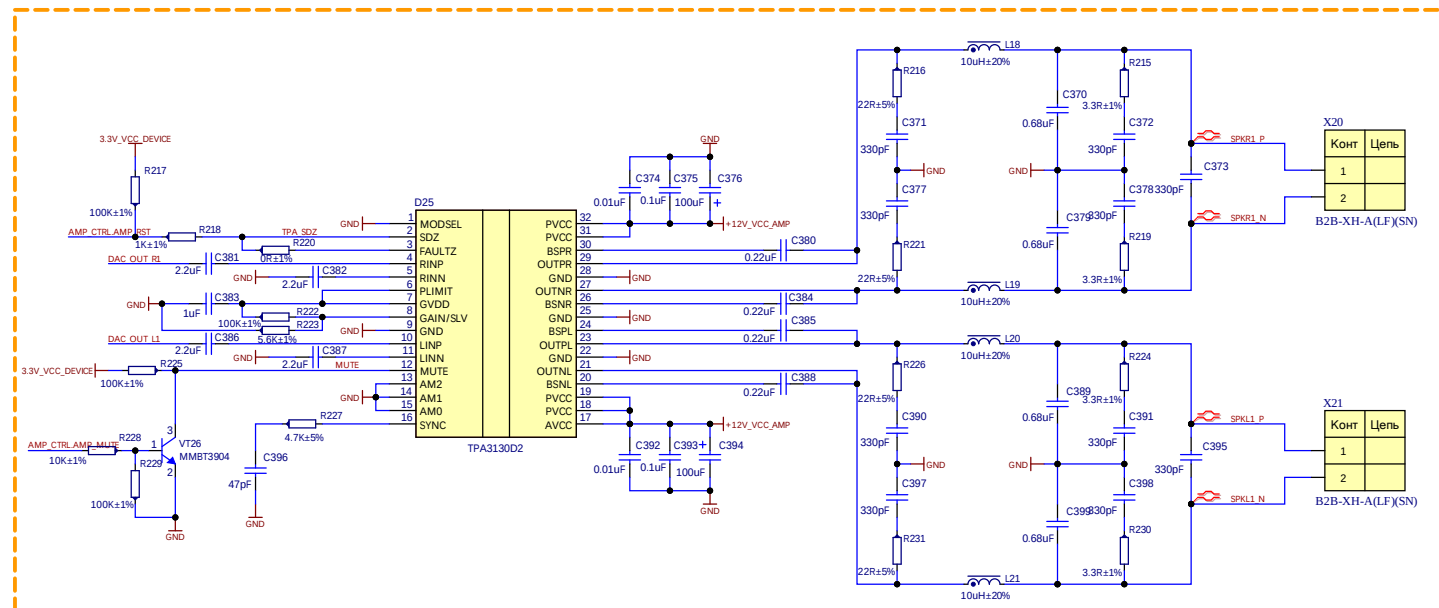


					ЮРТИ.466256.00233	Лист
Изм	Лист	№ докум.	Подп.	Дата		25

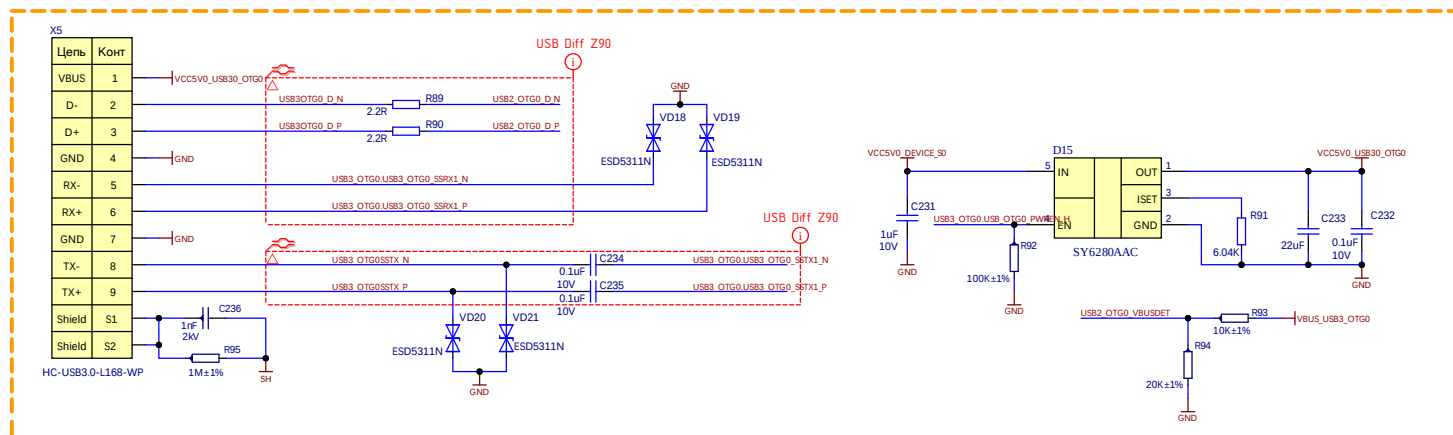
AMP PWR



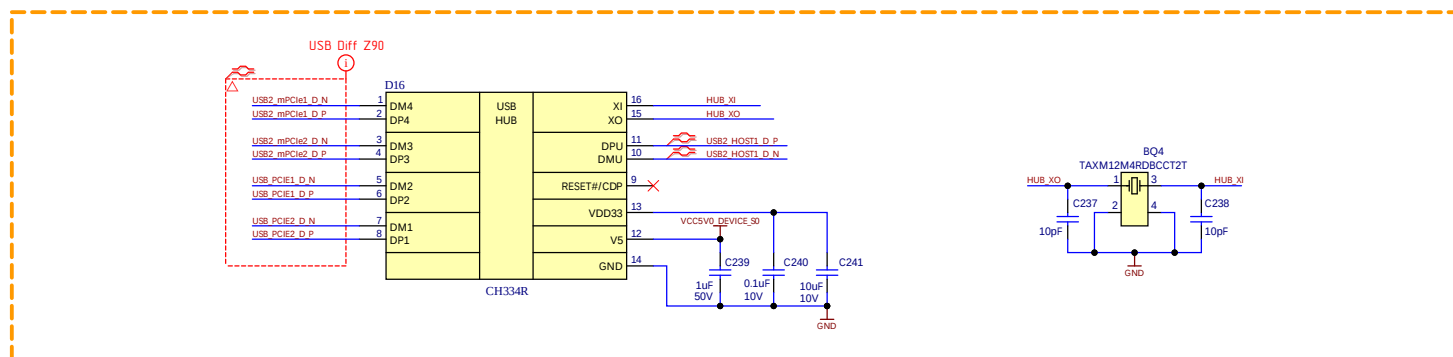
AMP SPK



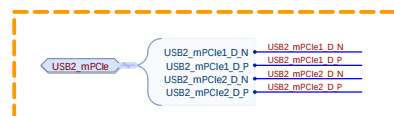
USB3.0 OTG - USB TYPE A



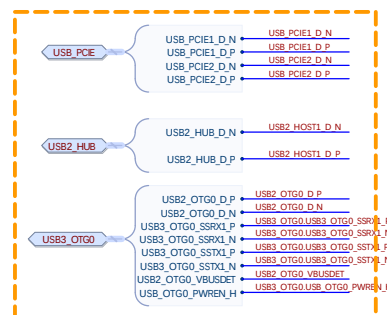
USB2.0 - HUB



Connection to INT_mPCIeX2_SIM



Connection to CPU_PCIE_USB_HDMI



Изм.	Лист	№ докум.	Подп.	Дата

A diagram showing the mapping of the **GPS_CTRL** pin to various peripheral control signals. The **GPS_CTRL** pin is connected to a block containing the following signals:

- GPS_MODE
- GPS_UART_RX
- GPS_UART_TX
- GPS_ON
- GPS_RST
- GPS_PPS

Each signal is connected to a specific peripheral control signal:

- GPS_MODE → GPS_CTRL.GPS_MODE
- GPS_UART_RX → GPS_CTRL.GPS_UART_RX
- GPS_UART_TX → GPS_CTRL.GPS_UART_TX
- GPS_ON → GPS_CTRL.GPS_ON
- GPS_RST → GPS_CTRL.GPS_RST
- GPS_PPS → GPS_CTRL.GPS_PPS

WL_SDIO

- WL_SDIO_D0
- WL_SDIO_D1
- WL_SDIO_D2
- WL_SDIO_D3
- WL_SDIO_CMD
- WL_SDIO_CLK

WL_BT_CTRL

- WL_EN
- WAKE_BT
- BT_EN_HOST

BT_UART

- BT_UART_BT_UART_CTS
- BT_UART_BT_UART_RX
- BT_UART_BT_UART_TX
- BT_UART_BT_UART_RTS

A diagram showing a blue arrow labeled "WL_BT_WAKE" pointing to a light blue box. From the right side of this box, two blue arrows branch out: the top one is labeled "WL_HOST_WAKE" and the bottom one is labeled "BT_WAKE".

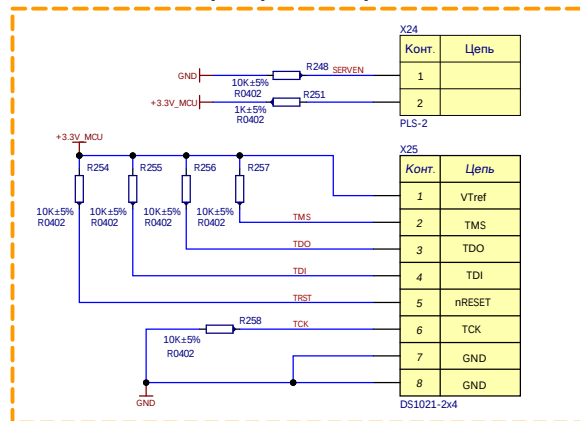
The diagram shows a blue arrow labeled "RTC CLK" pointing to a pin labeled "RTC_32KHZ". A blue line labeled "RTC 32KHZ" is connected to the "RTC_32KHZ" pin.

Конт	Цепь
1	
2	
3	

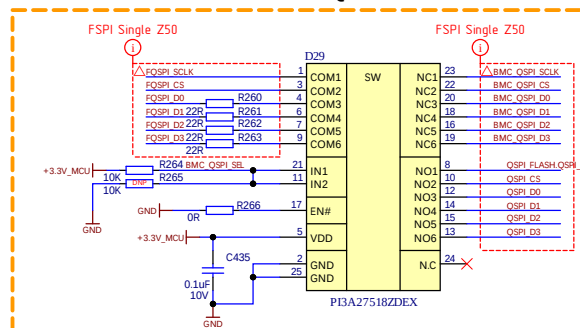
U.FL-R-SMT-1[80]

[illegible]

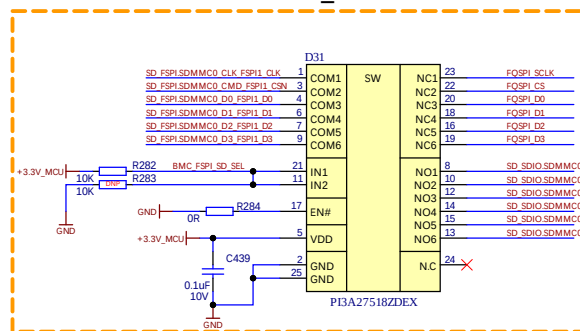
Блок программирования ВМС



MUX QSPI



MUX FSPI_SDCARD

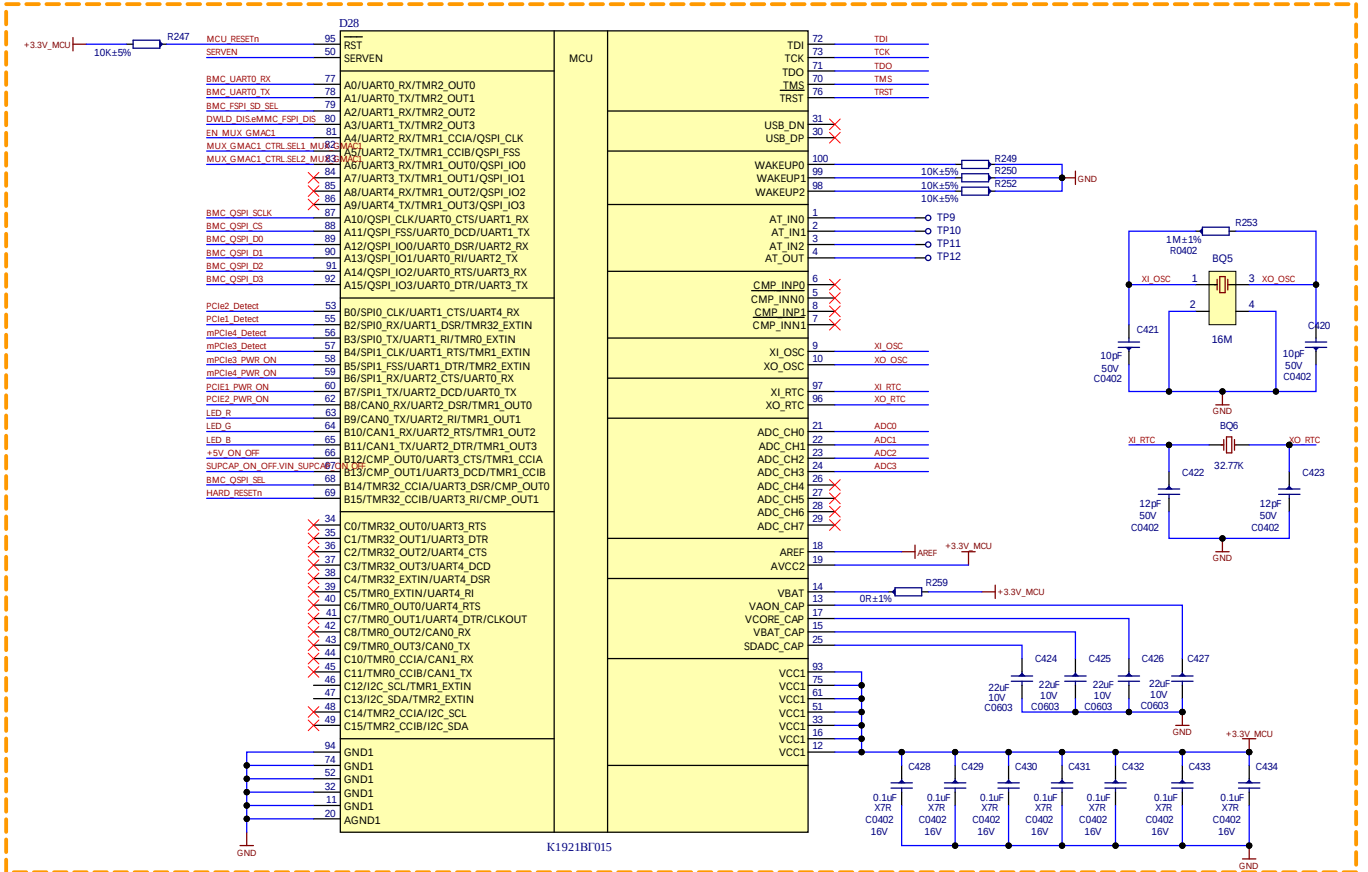


Connection to CPU GPIO

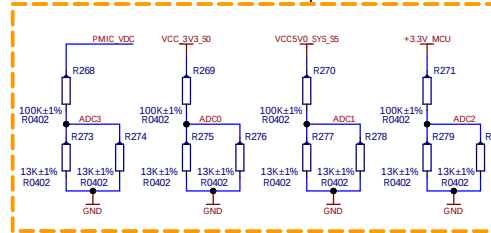
Connection to BMC

Connection to Flash eMMC MicroSD

BMC



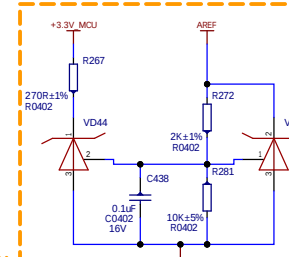
Блок АЦП



Connection to Peripheral

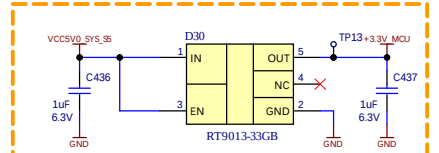
Connection to V-by-One

BMC AREF



Connection to BMC

Блок питания ВМС

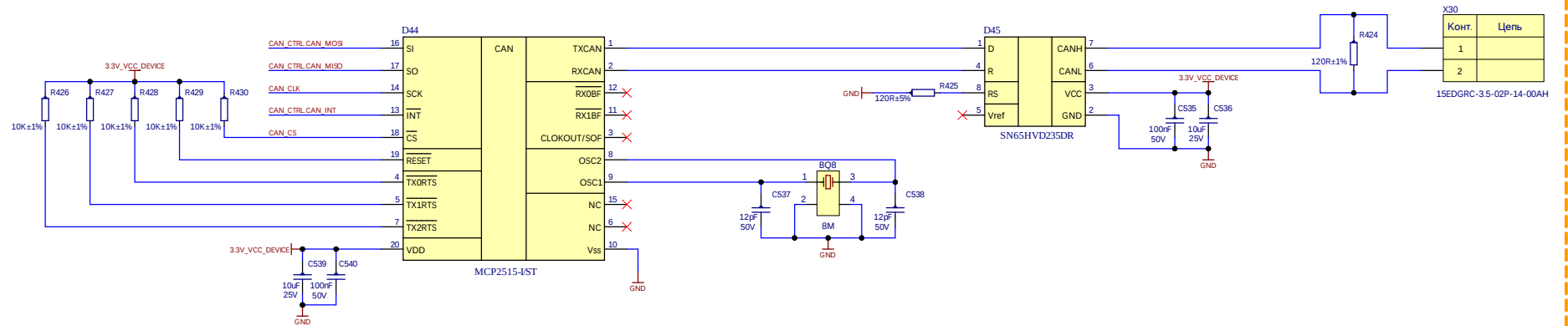


Connection to PWR PMIC/WIFI BT

Connection to BMC

Connection to BMC

CAN



Connection to USB

